Reading Multiple Regression Across Coordinated Questions

Jinsong Chen

Table of contents

Chapter box

Central question: How can one study answer marginal-association, conditional-association and prediction questions without treating their answers or supporting reasons as interchangeable?
Focal concepts: Common analytic records; partial slopes and indicators; coefficient covariance; support, form and influence; fitted and standardized quantities; held-out prediction error; separate arguments and their synthesis.
Main evidence: The 98 complete occupation records from `carData::Prestige`, followed by Project STAR's kindergarten reading comparison. Their different units and designs support different conclusions.
Scope: Chapter 7 supplies simple regression and Chapters 2–6 supply the statistical and R foundations. Chapter 1 supplies the reading approach; coordination is introduced locally and Chapter 1a remains optional. All nine sections, including STAR, form the analytical route. Appendices A–B retain optional algebra and technical reproduction. Reading comes before separate quality evaluation; construction is an auxiliary use.

Core roadmap. Section 1 separates three questions and their owed answers. Sections 2–5 develop the conditional-association question from coefficient meaning through estimation, checks, and selected fitted quantities. Section 6 asks the separate prediction question, and Section 7 reads, evaluates, and synthesizes the three occupation arguments. Section 8 then transfers the reading discipline to Project STAR's randomized design; Section 9 summarizes the route. The appendices are not required to understand the conceptual core; Appendix B is needed only to reproduce the complete all-fold and 200-resplit computations.

1 Three Questions in the Same Occupation Records

1.1 One study can owe several answers

A report may ask whether education and income move together, whether that relationship remains when other occupational characteristics are held constant, and whether those characteristics help predict income. These questions can inform one another, but answering one does not discharge the other two obligations.

We use a **chapter-authored retrospective teaching study**, not a published study with a newly discovered analysis plan. Its records are the *Prestige* data distributed with the carData package. The analyses and short report below are our teaching record; the package documentation supplies the data descriptions, not the claims written for this chapter.

Table 1
Three separately owed answers in the occupation teaching study.

Pair and role	Question	Estimand and owed answer
1. Contextual description	How does income vary linearly with education across the 98 retained occupations?	The finite-record marginal least-squares slope; a descriptive answer for these records
2. Primary conditional association	How does income differ with education at the same prestige, percentage of women, and type, within support?	The linear conditional-mean coefficient β_E under the declared working model; fitted b_E is its estimate from the 98 retained records
3. Secondary prediction analysis	Does the full additive learning rule predict eligible same-source occupation records with lower expected absolute error than the education-only rule?	Difference in expected absolute error of the two learning procedures at the declared training sizes; an internal validation answer

These are three question–answer pairs, each with **one argument with linked layers**. They are not three statistical strands of one argument: the report owes separate conclusions about marginal association, conditional association, and prediction. If several estimands were jointly necessary for *one* integrated conclusion, they could instead be parallel statistical strands. Merely printing several coefficients or diagnostics does not create either several questions or several strands.

1.2 Keep the records, time, and units visible

A row represents an **occupation**, not an individual worker. Income is the occupation's average annual income in 1971 Canadian dollars; we divide it by 1,000. Education is average years of education, and *women* is the percentage of women in the occupation. Prestige ratings were collected in a mid-1960s survey, whereas the income and education summaries come from 1971. Occupational type is blue-collar (*bc*), professional/managerial/ technical (*prof*), or white-collar (*wc*). These historical and aggregate measurements are not interchangeable with current individual earnings.

The earlier simple-regression analysis used all 102 occupations with education and income recorded. The declared full variable set leaves 98 complete records: athletes, newsboys, babysitters, and farmers have missing type. The retained type counts are 44, 31, and 23 respectively. We keep the original row order, which will also make the validation split reproducible.

```r
all_occ <- carData::Prestige
all_occ$income_k <- all_occ$income / 1000
all_occ$type <- factor(all_occ$type, levels = c("bc", "prof", "wc"))
vars <- c("income_k", "education", "prestige", "women", "type")
d <- all_occ[complete.cases(all_occ[, vars]), ]
c(original = nrow(all_occ), retained = nrow(d))
```

```
original retained 
     102       98 
```

```r
table(d$type)
```

```

  bc prof   wc 
  44   31   23 
```

We refit the education-only model on these **same 98 records** before comparing adjustment specifications. Moving from 102 to 98 changes which records are described. It must not be presented as an adjustment effect. Complete-case selection makes the computations comparable; it does not establish that the omitted occupations are irrelevant or that these records are a probability sample of Canadian occupations.

1.3 Declare the model question before reporting an interval

Pair 1 is finite-record description. Its slope is exactly calculable once the records and least-squares rule are specified.

Pair 2 has one intended estimand: the education coefficient β_E in the declared linear conditional-mean working model. The fitted coefficient b_E is its estimate from the 98 retained records. Although b_E is exactly calculable once the rows and fitting rule are fixed, that fact does not create a second finite-record estimand. A confidence interval additionally posits that occupation outcomes could vary independently around the stated linear conditional mean at the recorded predictors. This is not a claim that the dataset came from a verified random sample or that income was randomly assigned.

For the classical interval we additionally assume a common error variance; normal errors make its finite-sample t reference exact. Robust uncertainty relaxes the common-variance requirement under its own large-sample conditions. It does not create independence or linearity. We will check how credible and stable the working-model reading is before judging its answer.

A **best linear projection** can instead summarize a curved conditional relationship under a specified distribution of predictors. That is a different interpretation, not a rescue silently adopted after discovering curvature. The core uses the stated linear conditional-mean interpretation; Appendix B explains the alternative and its limitations.

Pair 3 will declare a different hypothetical repetition: train a rule on eligible, exchangeable same-source records, then predict a new record without using its outcome in training. Its justification cannot be supplied by the coefficient interval in Pair 2.

2 From Marginal to Conditional Comparisons

2.1 What a partial slope compares

Let e be education, p prestige, w percentage of women, and t type. The fitted additive equation is

$$\widehat{m}(e, p, w, t) = b_0 + b_E e + b_P p + b_W w + b_{\text{prof}} D_{\text{prof}} + b_{\text{wc}} D_{\text{wc}}.$$

An indicator D is 1 for its named category and 0 otherwise. Blue-collar is the reference type, with both indicators zero. This model has six coefficients, including its intercept.

Take two profiles with the same p, w, t and education differing by one year. Subtract their fitted means:

$$\begin{aligned} &\widehat{m}(e+1, p, w, t) - \widehat{m}(e, p, w, t) \\ &\quad = b_E(e+1) - b_E e = b_E. \end{aligned}$$

All the common terms cancel. Thus b_E compares modeled occupation-level income at the **same prestige, percentage of women, and type**. A two-year difference corresponds to $2b_E$ under this additive model.

"Holding constant" describes this mathematical comparison. It does not say that education was changed experimentally, that two observed occupations match exactly on every other characteristic, or that unmeasured differences are controlled. An additive slope also assumes the same education comparison across the included predictor values. Allowing it to vary with another predictor is the interaction question taken up later.

2.2 Why the marginal slope can differ

A short known-process example makes the distinction visible. Suppose

$$Y = 10 + 2X + 5Z + \varepsilon, \qquad E(\varepsilon \mid X, Z) = 0,$$

with finite variances and $\mathrm{Cov}(X, Z)/\operatorname{Var}(X) = 0.65$. The conditional X coefficient is 2. The marginal linear slope is

$$\frac{\mathrm{Cov}(X, Y)}{\operatorname{Var}(X)} = 2 + 5\frac{\mathrm{Cov}(X, Z)}{\operatorname{Var}(X)} = 2 + 5(0.65) = 5.25.$$

The marginal comparison incorporates how Z moves with X in this specified process. The conditional comparison holds Z fixed. Neither arithmetic result alone says what would happen if a researcher intervened on X. Calling 5.25 "biased" requires naming an estimand for which it is being used incorrectly; as a marginal linear summary, 5.25 is the right answer here. Appendix A retains the derivation and simulation.

More generally, adding predictors can change the comparison, but it does not **always** change the scientific estimand. In a randomized study, appropriately specified covariate adjustment may estimate the same declared average assignment effect more precisely. Releveling a factor changes coefficient coordinates without changing fitted values. A different uncertainty estimator may change the interval without changing the fitted coefficient.

2.3 A small indicator bridge

The full occupation fit, rounded for substitution, is

$$\begin{aligned}\widehat{\mathrm{income}}_k = 0.420524 &+ 0.116434e + 0.140605p - 0.053505w \\ &+ 0.324193D_{\mathrm{prof}} + 0.235331D_{\mathrm{wc}}.\end{aligned}$$

Consider a white-collar profile with education 11, prestige 45, and women 50. Here $D_{\mathrm{prof}} = 0$ and $D_{\mathrm{wc}} = 1$:

$$\begin{aligned}\widehat{\mathrm{income}}_k &= 0.420524 + 0.116434(11) + 0.140605(45) \\ &\quad - 0.053505(50) + 0 + 0.235331 \\ &\approx 5.589.\end{aligned}$$

That is a fitted mean of about 5,589 dollars on the historical income scale, not a guarantee for any occupation or worker. Can these three predictor values occur together within the observed white-collar covariate cloud? Separate minimum and maximum values cannot answer that question.

Here is a checkable **joint-support witness**. A convex combination is a weighted average with nonnegative weights summing to one. Four retained white-collar records can be combined to recover education 11, prestige 45, and women 50 simultaneously. The source spelling *receptionsts* is retained in the code so that the record can be located exactly.

Use the supplied weights to check the three weighted averages. The order of the weights matches the four named records. `weighted.mean()` performs the same weighted-average calculation for each variable; finding these weights is not a prerequisite for interpreting their evidence.

```
support_names <- c("radio.tv.announcers", "tellers.cashiers",
                   "receptionsts", "sales.supervisors")
support_rows <- d[support_names, c("education", "prestige", "women")]
support_weights <- c(0.245243736035, 0.230972611567,
                     0.226143656937, 0.297639995461)
d[support_names, "type"]
```

```
[1] wc wc wc wc
Levels: bc prof wc
```

```
sum(support_weights)
```

```
[1] 1
```

```
c(education = weighted.mean(support_rows$education, support_weights),
  prestige = weighted.mean(support_rows$prestige, support_weights),
  women = weighted.mean(support_rows$women, support_weights))
```

```
education  prestige     women 
       11        45        50 
```

Table 2
Four observed white-collar records recover the declared profile.

Occupation	education	prestige	women	Weight
Radio/TV announcers	12.71	57.6	11.15	0.2452
Tellers/cashiers	10.64	42.3	91.76	0.2310
Receptionists	11.04	38.7	92.86	0.2261
Sales supervisors	9.84	41.5	17.04	0.2976

The supplied weights are all positive and sum to one; their weighted coordinates recover $(11, 45, 50)$ within 10^{-10} using the displayed precision. The table rounds weights for reading. This establishes membership in the joint covariate hull **within white-collar type**, using all three continuous predictors in the full model. It does not show that many occupations lie near this profile, that observed occupations match exactly, or that a causal comparison is supported. Local data may still be sparse.

The intercept refers to zero education, prestige, and women in the reference type. It helps construct the fitted surface; it need not describe a plausible occupation. Likewise, one type coefficient is not "the effect of occupational type" as a whole. Appendix A retains the full factor, group-mean, contrast-covariance, and classical ANOVA bridge. That material extends the compact indicator calculation without becoming the main route.

3 Estimate the Conditional Association

3.1 Compare specifications on the same rows

```
m0 <- lm(income_k ~ education, data = d)
m1 <- lm(income_k ~ education + prestige, data = d)
m2 <- lm(income_k ~ education + prestige + women + type, data = d)
coef(m2)
```

```
(Intercept)   education    prestige       women    typeprof
     0.4205      0.1164      0.1406     -0.0535      0.3242
     typewc
     0.2353
```

Table 3

The model sequence separates sample composition from adjustment.

Records and specification	Education slope	In-sample R^2	Role here
Earlier 102, education only	0.899	.334	Earlier result; not the adjustment baseline
Same 98, education only	0.883	.330	Pair 1's marginal description
Same 98, education and prestige	-0.218	.500	Intermediate comparison
Same 98, full declared predictors	0.116	.636	Pair 2's estimate of β_E

Note. Slopes are thousand 1971 Canadian dollars per education year. These are fitted summaries, not causal effects. No sampling interval is attached to Pair 1's finite-record estimand.

Education and prestige are highly correlated here, $r = .866$. Once prestige is included, the education comparison uses variation not linearly accounted for by prestige. Adding women and type changes the conditioning set again. The movement from .883 to -.218 to .116 is not a narrative in which a true effect is successively "explained away." It is a sequence of named comparisons in a small observational record.

The increase in R^2 describes training-data fit. Adding predictors cannot increase the least-squares residual sum of squares on the same rows when the models are nested. Better training fit does not establish better prediction on records left out of fitting.

3.2 Partial regression gives the same coefficient

Multiple regression still uses the covariance reasoning from simple regression. First regress education on the other predictors and retain its residuals $r_{E,i}$. Do the same for income, obtaining $r_{Y,i}$. Both auxiliary regressions include an intercept. Then

$$b_E = \frac{\sum_i r_{E,i} r_{Y,i}}{\sum_i r_{E,i}^2}.$$

The numerator measures remaining linear co-variation; the denominator measures remaining education variation. This formula is the partial regression identity, not a new estimation method with a new question.

```r
r_E <- residuals(lm(education ~ prestige + women + type, data = d))
r_Y <- residuals(lm(income_k ~ prestige + women + type, data = d))
sum(r_E * r_Y) / sum(r_E^2)
```

```
[1] 0.1164
```

```r
cor(r_E, r_Y)
```

```
[1] 0.04406
```

The slope is 0.1164 and the partial correlation is .0441. The latter is unitless; it is not the slope expressed in dollars. Both summarize the same residualized comparison in different units.

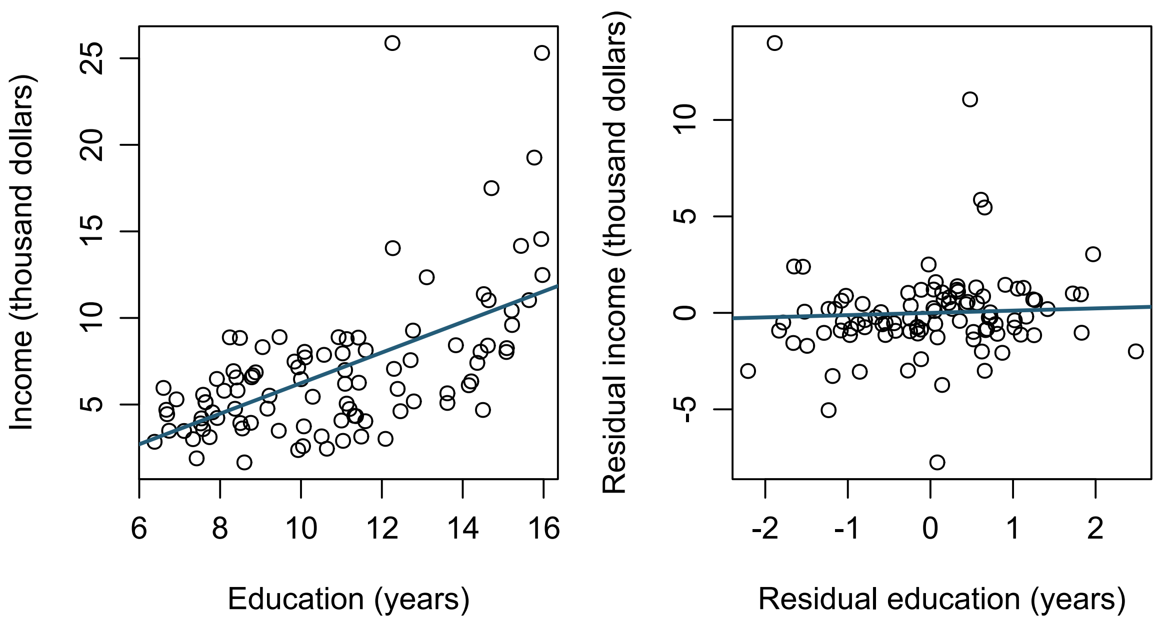

Figure 1: Marginal and residualized income–education comparisons in the same 98 occupations.

Residualization does not remove all substantive differences between occupations. It removes the **fitted linear components** of the named variables. Appendix A develops the matrix version.

3.3 Read a coefficient interval under its stated model

The hypothetical linear conditional-mean model introduced in Section 1 has six estimated coefficients, so its residual degrees of freedom are $98 - 6 = 92$. For education the classical calculation is

$$
\begin{aligned}
b_E \pm t_{.975,92} SE(b_E) &= 0.1164 \pm 1.9861(0.2752) \\
&\approx [-0.430,\ 0.663].
\end{aligned}
$$

The result is compatible, under those assumptions, with negative and positive conditional slopes on this scale. It does not demonstrate that education is unimportant, that the slope equals zero, or that adjustment has removed an education effect. No substantive equivalence margin was declared, and the interval includes differences of several hundred dollars per year of education. The data and model therefore leave the direction and practically relevant magnitude unresolved.

```r
confint(m2, "education")
```

```
            2.5 % 97.5 %
education -0.4302 0.6631
```

```r
hc3_covariance <- sandwich::vcovHC(m2, type = "HC3")
hc3_se <- sqrt(hc3_covariance["education", "education"])
coef(m2)["education"] + c(-1, 1) * qt(.975, 92) * hc3_se
```

```
[1] -0.6429  0.8758
```

HC3 gives SE 0.3823 and an approximate interval $[-0.643, 0.876]$ using the displayed t_{92} convention. It allows heteroskedasticity under independent-observation and other regularity conditions; it is not an exact small-sample guarantee. Both intervals leave broad uncertainty. Neither accounts for unmodeled dependence, selection, measurement error, or a wrong conditional-mean form.

If a test is reported, its target belongs with this inference: the two-sided education-coefficient test addresses $H_0 : \beta_E = 0$ in the stated full model, not "education has no effect." Its classical $t = 0.423$ and $p = .673$ do not prove the null. We do not use this test to choose which of the three questions deserves an answer.

4 Check Support, Form, Influence, and Precision

Checking asks what the model leaves unexplained and what follows for the answer. It is not a ritual in which each graph receives a pass or fail. Residuals, support, coefficient precision, and sensitivity each answer a different diagnostic question.

4.1 Read residual, scale, tail, and influence views together

A residual is an observed deviation from a fitted value, not the unobserved population error. A standardized residual uses a common residual-scale estimate; an externally studentized residual uses a scale estimate with that observation omitted. Leverage measures an unusual predictor configuration. Influence combines predictor and outcome information (Cook, 1977).

```r
ch8_diag <- data.frame(
  occupation = rownames(d), fitted = fitted(m2),
  residual = residuals(m2), standardized = rstandard(m2),
  studentized = rstudent(m2), leverage = hatvalues(m2),
  cooks = cooks.distance(m2)
)
ch8_top <- order(ch8_diag$cooks, decreasing = TRUE)[1:5]
```

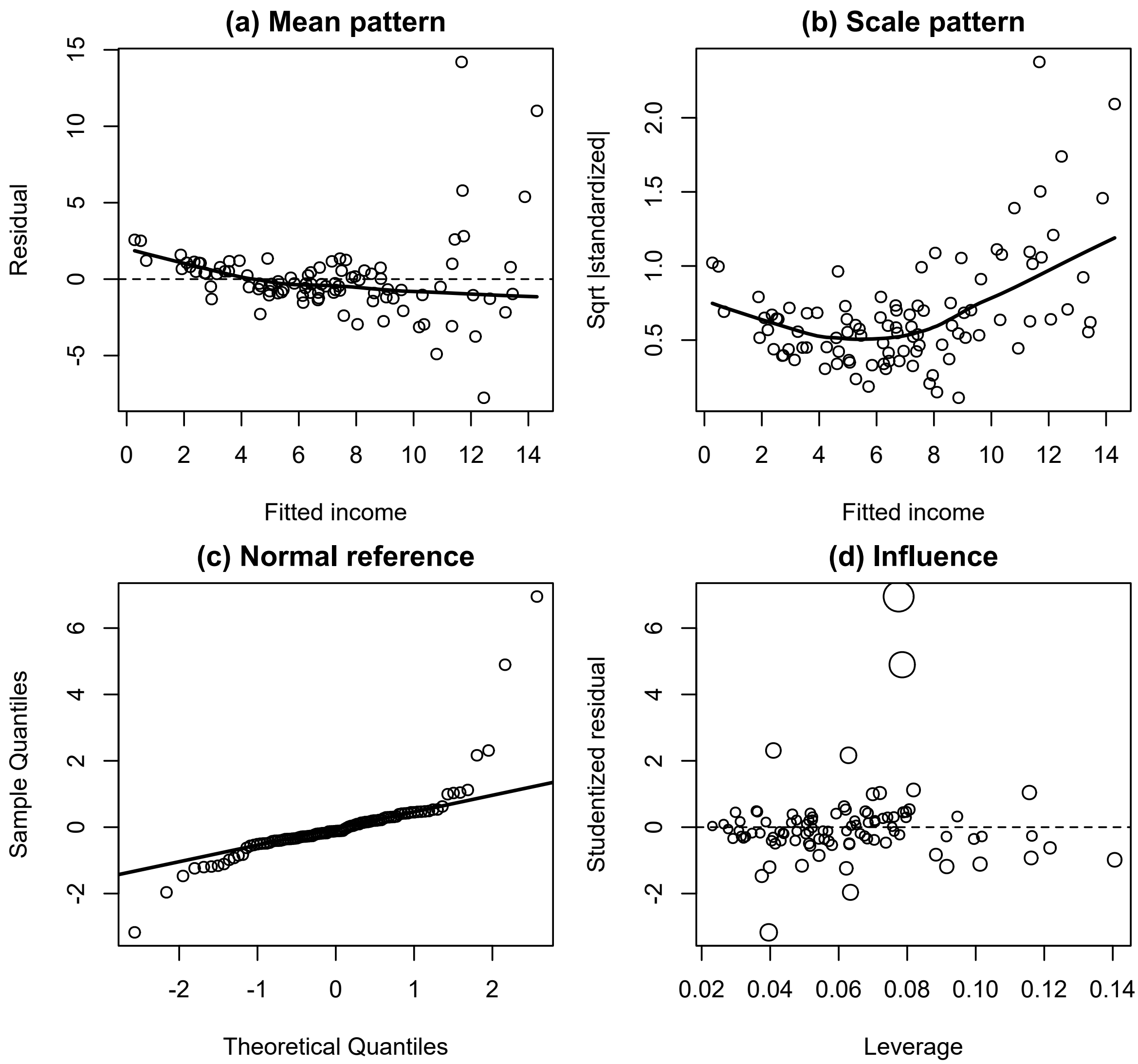

Figure 2: Four complementary diagnostic views for the full additive model.

The smooth and changing spread challenge a constant conditional-mean and variance description. The right tail matters to exact small-sample normal-theory inference. Neither feature prevents calculation of OLS. Large residual, high leverage, and high influence are different features; none is an automatic record-invalidity verdict. No residual graph can establish that occupations are independent draws from a wider population.

4.2 Inspect joint support and coefficient precision

Joint support concerns predictor combinations, not merely separate minimum and maximum values. The convex-combination witness in Section 2 establishes support for the focal white-collar profile across all three continuous predictors. The marginal ranges below ask a different, coarser question: how far does each variable extend within each recorded type?

Table 4
Predictor ranges within each recorded occupation type.

	n	Education	Prestige	Women
bc	44	6.38–10.93	17.3–54.9	0.00–90.67
prof	31	11.09–15.97	53.8–87.2	0.58–96.12
wc	23	9.17–12.79	26.5–67.5	3.16–97.51

Blue-collar education ends at 10.93 years, below the professional minimum of 11.09. A same-education comparison between those types necessarily crosses that gap. A model can return both predictions there; software output does not create observations. Lack of support limits what can be learned or interpreted about a defined comparison rather than making the estimand cease to exist.

A conditional grid can also avoid pretending that an unrestricted rectangle is supported. For *m1*, hold prestige at 30, 50, or 70 and plot education only over values observed within 7.5 prestige points of each target. The width is an explicit display convention, not a formal support guarantee.

```
ch8_grid <- data.frame()
for (p0 in c(30, 50, 70)) {
  nearby <- d[abs(d$prestige - p0) <= 7.5, ]
  one_grid <- data.frame(
    education = seq(min(nearby$education), max(nearby$education),
                    length.out = 40),
    prestige = p0, Profile = paste("Prestige", p0)
  )
  ch8_grid <- rbind(ch8_grid, one_grid)
}
ch8_grid$fit <- predict(m1, newdata = ch8_grid)
```

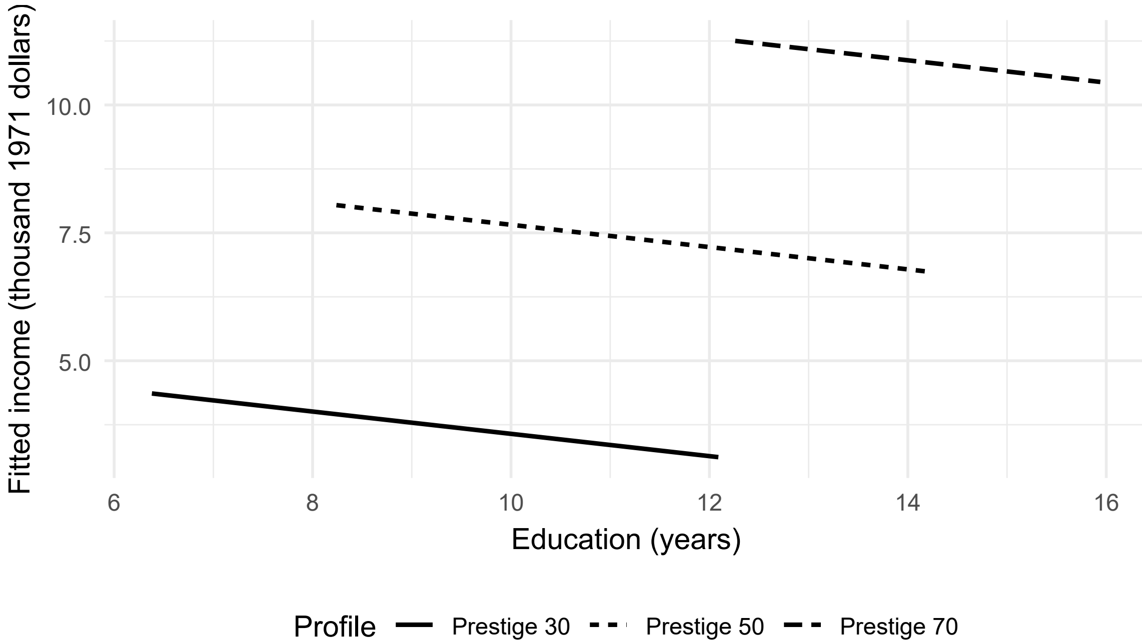

Figure 3: Neighborhood-restricted fitted profiles from the education-and-prestige model.

The parallel lines share *m1*'s education slope. This display conditions on prestige only, not on the full set for Pair 2. It teaches a transparent plotting choice; it does not prove support for the full-model profile.

Education's variance inflation factor makes the collinearity calculation checkable:

$$VIF_E = \frac{1}{1 - R^2_{E|P,W,T}} = \frac{1}{1 - .876495} \approx 8.097.$$

Under the classical model, holding residual variance and total education variation fixed, its coefficient SE is multiplied by $\sqrt{8.097} \approx 2.846$ relative to an orthogonal design. This is a comparison of variance formulas, not a claim that deleting other predictors will divide the actual SE by 2.846. Deletion changes the comparison and can change residual variance. VIF is not an automatic deletion threshold, and near collinearity does not by itself create bias under the maintained zero-conditional-mean model.

4.3 Use targeted sensitivities to bound the answer

A component-plus-residual plot adds the fitted prestige contribution back to each residual, $e_i + \hat{\beta}_P P_i$. The smooth bends upward at high prestige relative to the straight fitted component.

That pattern motivates a declared form sensitivity; it does not prove that one chosen curve is the true process.

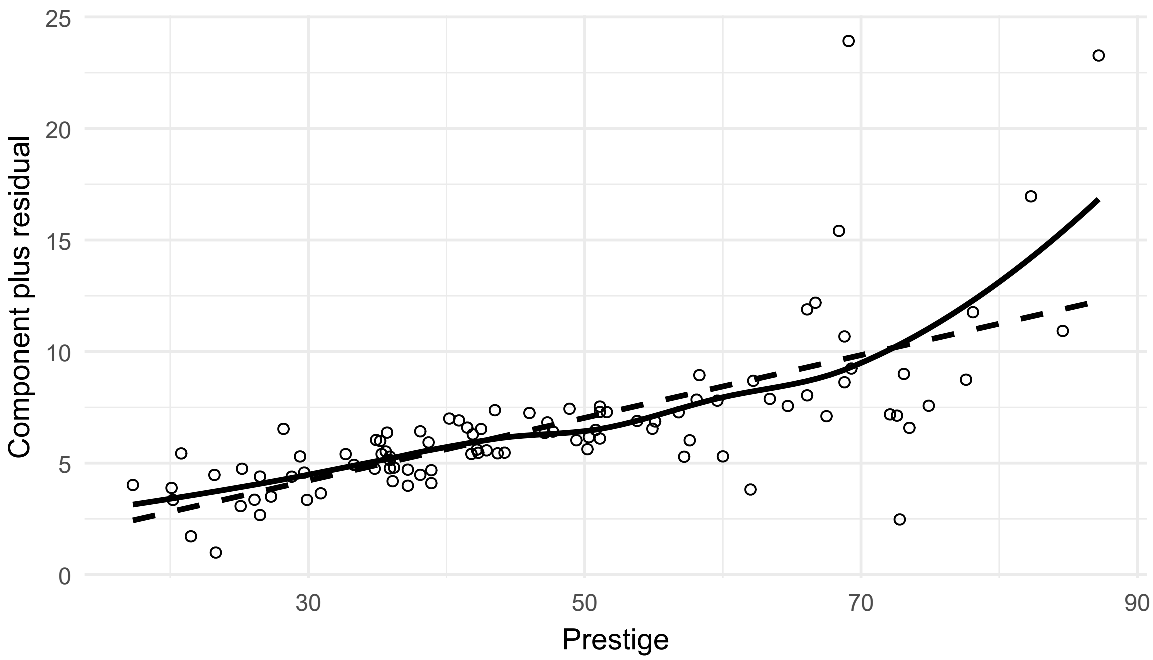

Figure 4: The prestige component plus residual in the full additive model.

```r
ch8_curve <- update(m2, . ~ . + I(prestige^2))
anova(m2, ch8_curve)
```

```
Analysis of Variance Table

Model 1: income_k ~ education + prestige + women + type
Model 2: income_k ~ education + prestige + women + type + I(prestige^2)
  Res.Df RSS Df Sum of Sq    F Pr(>F)
1     92 631
2     91 587  1      43.5 6.74  0.011 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
c(full_education = unname(coef(m2)["education"]),
  curve_education = unname(coef(ch8_curve)["education"]))
```

```
 full_education curve_education
         0.1164         -0.0298
```

The classical nested comparison gives $F(1, 91) = 6.736$ and $p = .011$; education changes from 0.116434 to -0.029800. In the enlarged equation, the prestige slope depends on prestige through $\beta_P + 2\beta_{P^2}P$. This is a focused sensitivity for Pair 2, not a validation-tuned replacement for the frozen prediction rule in Pair 3.

The record for general managers has the largest Cook's distance, about 0.447. Its high income relative to the model is not proof of error. A transparent deletion refit asks how dependent the answer is on that record:

```r
ch8_without_top <- lm(
  income_k ~ education + prestige + women + type,
  data = d[-ch8_top[1], ]
)
c(full_education = unname(coef(m2)["education"]),
  omit_one_education = unname(coef(ch8_without_top)["education"]))
```

```
    full_education omit_one_education
            0.1164             0.4372
```

The education estimate moves to 0.437189. Preserve the full-data estimate and disclose the sensitivity. Verify coding, measurement, and substantive distinctiveness before considering exclusion; a more attractive p value is not an exclusion rule. The refit is a diagnostic perturbation, not a new research question or evidence that the reduced sample is preferable.

Table 5
Checks change the bounded answer rather than producing one omnibus pass/fail score.

Check	What is visible here	Consequence for the answer
Functional form	Adding a prestige-squared term changes the education slope to about -0.030	Treat the additive form as a working approximation; do not silently reinterpret it as an exact conditional law
Spread	HC3 raises the education SE from .275 to .382	Show sensitivity; a robust SE does not repair all model problems
Influence	Removing the record for general managers, which has the largest Cook's distance, gives an education slope about .437	Inspect the record and disclose sensitivity; do not delete it merely to obtain a preferred result
Collinearity	Education's auxiliary R^2 is .8765	Little education variation remains after the other predictors; separate conditional coefficients can be imprecise
Joint support	Professional and blue-collar education ranges do not overlap	Cross-type comparisons at common covariates may depend heavily on extrapolation
Dependence and scope	Occupational aggregates have shared economic and classification contexts	Independence and same-source transport are assumptions, not verified by a residual plot

Appendix B retains the full Cook's distance and deletion accounting, manual HC3 covariance reproduction, generalized VIF, changed-outcome models, and model-comparison criteria. These are technical extensions of checks already interpretable here, not prerequisites hidden outside the numbered route.

5 Move from Coefficients to Fitted and Standardized Quantities

The profile and standardization calculations in this section are **didactic extensions of the fitted model**, not extra question-facing answers owed by the occupation study. Pair 2 remains about β_E. A report that promoted one of these quantities to its own substantive conclusion would need to state the corresponding question, estimand, justification, and bounds.

5.1 A fitted mean is not prediction performance

Start with a complete input row and ask R for the two interval types. `interval = "confidence"` concerns the model mean at that profile; `interval = "prediction"` concerns one new outcome at the same profile. Both use the full fitted model and its stated classical assumptions.

```
profile <- data.frame(education = 11, prestige = 45, women = 50,
                      type = factor("wc", levels = levels(d$type)))
mean_check <- predict(m2, newdata = profile, interval = "confidence")
new_check <- predict(m2, newdata = profile, interval = "prediction")
mean_check
```

```
    fit   lwr   upr
1 5.589 4.486 6.691
```

```
new_check
```

```
    fit    lwr   upr
1 5.589 0.2722 10.91
```

For the white-collar profile in Section 2, collect the coefficient multipliers in $x_* = (1, 11, 45, 50, 0, 1)'$. Their order is intercept, education, prestige, women, professional indicator, and white-collar indicator. The fitted mean and its estimated classical variance are below. A prime denotes transpose (`t()` in R); `vcov(m2)` is the estimated covariance matrix of the fitted coefficients, not the covariance matrix of the observed variables.

$$\widehat{m}_* = x_*'\hat{\beta}, \qquad \widehat{\mathrm{Var}}(\widehat{m}_*) = x_*'\widehat{V}x_*, \quad \widehat{V} = \mathrm{vcov}(m2).$$

The variance includes the coefficient variances **and their covariances**. For example, two terms $a\hat{\beta}_j + b\hat{\beta}_k$ contribute $a^2\widehat{\mathrm{Var}}(\hat{\beta}_j) + b^2\widehat{\mathrm{Var}}(\hat{\beta}_k) + 2ab\widehat{\mathrm{Cov}}(\hat{\beta}_j, \hat{\beta}_k)$. Adding coefficient SEs, or ignoring covariance, is not the same calculation. Take the square root only after calculating the variance.

```r
x_star <- c(1, 11, 45, 50, 0, 1)
V <- vcov(m2)
profile_fit <- sum(x_star * coef(m2))
profile_var <- drop(t(x_star) %*% V %*% x_star)
s2 <- sum(residuals(m2)^2) / df.residual(m2)
critical <- qt(.975, df.residual(m2))
profile_mean_ci <- profile_fit +
  c(-1, 1) * critical * sqrt(profile_var)
profile_new_pi <- profile_fit +
  c(-1, 1) * critical * sqrt(s2 + profile_var)
rbind(fitted_mean = profile_mean_ci, new_case = profile_new_pi)
```

```
              [,1]   [,2]
fitted_mean 4.4865  6.691
new_case    0.2722 10.905
```

The fitted mean is 5.588634; its estimated variance is 0.307962 and SE is $\sqrt{0.307962} = 0.554943$, in thousand-dollar units for the SE. Variance has squared units. The classical fitted-mean interval is therefore $[4.486, 6.691]$ thousand dollars.

For **one new outcome**, its own error variation adds $s^2 = 6.857497$: the SE is $\sqrt{s^2 + x_*'\widehat{V}x_*}$, giving the much wider interval $[0.272, 10.905]$. This calculation uses the independent normal common-variance conditional-mean model; it is not a promise established by the diagnostics. Replacing $\widehat{V}$ with HC3 alone does not estimate an arbitrary new outcome's conditional error distribution.

Neither interval tells us how accurately a learning rule predicts held-out records on average. That requires Pair 3.

5.2 Standardize over a named distribution

A conditional fitted value uses one specified profile. A standardized fitted value averages across a named distribution. To see the operation, copy the 98 observed profiles, change their type to white-collar, predict each fitted value, then take the mean. Each continuous profile has equal weight.

```r
white_profiles <- d
white_profiles$type <- factor("wc", levels = levels(d$type))
white_predictions <- predict(m2, newdata = white_profiles)
mean(white_predictions)
```

```
[1] 7.016
```

This is an average of model predictions after a specified profile change, not the observed white-collar group mean. For type g and fixed nonnegative weights w_i summing to one,

$$\hat{\mu}_g^{\text{std}} = \sum_i w_i x_{ig}' \hat{\beta} = a_g' \hat{\beta}, \qquad a_g = \sum_i w_i x_{ig}.$$

Here x_{ig} is the design vector after assigning type g to observed continuous profile i. With empirical weights $w_i = 1/98$, this is standardization over these 98 profiles, not over a national population. Treating those profiles and weights as fixed gives estimated variance $a_g' \widehat{V} a_g$.

The supplied loop repeats the same change–predict–average operation for each type. `model.matrix()` gives the six coefficient multipliers for every changed profile; `colMeans()` averages each multiplier to obtain a_g. The variance calculation includes covariance between all predictions through their shared fitted coefficients. Averaging individual interval endpoints does not give an interval for this standardized mean.

```r
ch8_std <- data.frame(Type = levels(d$type), Estimate = NA_real_,
                      SE = NA_real_)
V <- vcov(m2)
for (j in seq_len(nrow(ch8_std))) {
  changed_profiles <- d
  changed_profiles$type <- factor(
    ch8_std$Type[j], levels = levels(d$type)
  )
  ch8_std$Estimate[j] <- mean(predict(m2, newdata = changed_profiles))
  X <- model.matrix(~ education + prestige + women + type,
                    data = changed_profiles)
  a <- colMeans(X)
  ch8_std$SE[j] <- sqrt(drop(t(a) %*% V %*% a))
}
critical <- qt(.975, df.residual(m2))
ch8_std$Lower <- ch8_std$Estimate - critical * ch8_std$SE
ch8_std$Upper <- ch8_std$Estimate + critical * ch8_std$SE
```

Table 6

Empirical-profile standardization under the full additive model.

Type	Estimate	SE	Lower	Upper
bc	6.781	0.694	5.402	8.160
prof	7.105	0.901	5.316	8.895
wc	7.016	0.618	5.790	8.243

Because *m2* is linear and additive in the continuous predictors, its average fitted value equals its fitted value at their means. This is an algebraic special case: with squared terms, interactions, or a nonlinear response function, prediction at means need not equal mean prediction. Constant additive type contrasts remain unchanged by common weights, although absolute standardized levels can change.

The overall education mean is 10.795 years. Relabelling every profile professional or white-collar creates some unsupported combinations. Averaging them does not solve that problem. The white-collar profile used above is a different, locally supported conditional quantity; it is not a three-type comparison. The intervals condition on the empirical profiles and use the classical covariance. Estimating a new population's covariate distribution would add uncertainty not included here.

Standardization does not identify a causal effect by itself and does not validate out-of-sample performance. Pair 3 requires a learning rule and held-out outcomes.

6 Ask and Answer the Separate Prediction Question

6.1 Define the learning rule, new case, and loss

A learning rule says how training data become a fitted prediction equation. Our two rules are fixed: fit education-only OLS, or fit the full additive OLS equation with education, prestige, women, and type. Both use the same outcome and records. OLS **fits** by minimizing squared training errors; we separately choose **mean absolute error (MAE)** to evaluate the resulting predictions in interpretable income units.

A prediction opportunity here means a hypothetical same-source occupation with all declared predictors available before its income outcome is used. It does not mean predicting future earnings from information that would only become available afterward. The records do not establish that this hypothetical exchangeability assumption holds for new countries, periods, individual workers, or occupational types absent from the dictionary.

For held-out outcomes y_i and predictions $\hat{y}_i^{(-)}$ fitted without those outcomes,

$$MAE = \frac{1}{n_{\text{test}}} \sum_{i \in \text{test}} \left| y_i - \hat{y}_i^{(-)} \right|.$$

The superscript reminds us that the case was held out of fitting. A smaller number means smaller average absolute prediction error on the evaluated cases. It does not say that every case was predicted better.

6.2 Keep held-out outcomes out of training

We use shared ten-fold cross-validation. The 98 records are divided into ten folds; each fold takes a turn as the held-out set. Fit both equations on the other records, predict the held-out cases, then combine the 98 case-level errors.

Table 7
The internal-validation protocol was fixed before this teaching comparison.

Decision	Frozen teaching protocol	Why it matters
Candidates	Education-only and the declared full additive OLS rule	No search for the best-looking model after validation
Folds	Shared balanced random ten-fold assignment, seed 80806	Both rules face the same held-out cases
Training sizes	Eight fits use 88 records; two use 89	Risk concerns those training sizes, not a model trained on all 98
Preparation	Same 98 complete records; fixed income/1,000 conversion and type dictionary	No outcome-informed preprocessing; selection still limits scope
Estimability	Every training set must contain all three types and have full column rank	Stop rather than silently substitute a different rule
Summary	Pool absolute errors over all 98 held-out cases	Each case receives equal weight; folds of different size do not
Sensitivity	200 new resplits of the same records, seed 80807	Describes sensitivity to splitting, not new independent studies

This is a retrospective teaching reanalysis of familiar historical data. "Fixed before this comparison" is not a claim of historical preregistration or that analysts had never seen the outcomes. The candidates are not revised after inspecting the primary errors; the association sensitivities in Section 4 are not covert additions to the validated candidate set.

```r
set.seed(80806)
fold <- sample(rep(seq_len(10), length.out = nrow(d)))
held_fold <- fold[1]
train <- d[fold != held_fold, ]
held <- d[fold == held_fold, ]
fold_simple <- lm(income_k ~ education, data = train)
fold_full <- lm(income_k ~ education + prestige + women + type,
                data = train)
table(train$type)
```

```

  bc prof   wc
  37   29   23
```

```r
c(training = nrow(train), held_out = nrow(held),
  simple_rank = fold_simple$rank, full_rank = fold_full$rank)
```

```
   training    held_out simple_rank   full_rank
         89           9           2           6
```

```r
first_case <- d[1, , drop = FALSE]
first_predictions <- c(
  Education = unname(predict(fold_simple, first_case)),
  Full = unname(predict(fold_full, first_case))
)
rbind(prediction = first_predictions,
      absolute_error = abs(first_predictions - first_case$income_k))
```

```
               Education    Full
prediction         8.946 11.5948
absolute_error     3.405  0.7562
```

The check should show all three types, 89 training records and nine held-out records. The fitted ranks are 2 and 6, the numbers of coefficient columns. Proceed with the declared comparison only when its training data can estimate both rules. The full reproduction in Appendix B checks this for every fold.

This is one fold of the frozen protocol, with 89 training records and 9 held-out records. Government administrators, the first source record, is in fold 9 and was excluded from both fits. The displayed code predicts that record; repeat the same train/predict/error steps across all ten folds to obtain the all-98 comparison below. One record's error is not that summary.

Appendix B provides the complete ten-fold function and 200-resplit code as optional computational reproduction. The visible one-fold example needs only the preceding core data definitions, not that helper. No held-out outcome is used to estimate a coefficient. If we introduced learned scaling, imputation, or variable selection, each step would have to be fitted using training data only. A fixed unit conversion such as dollars to thousands is not an estimated preprocessing step.

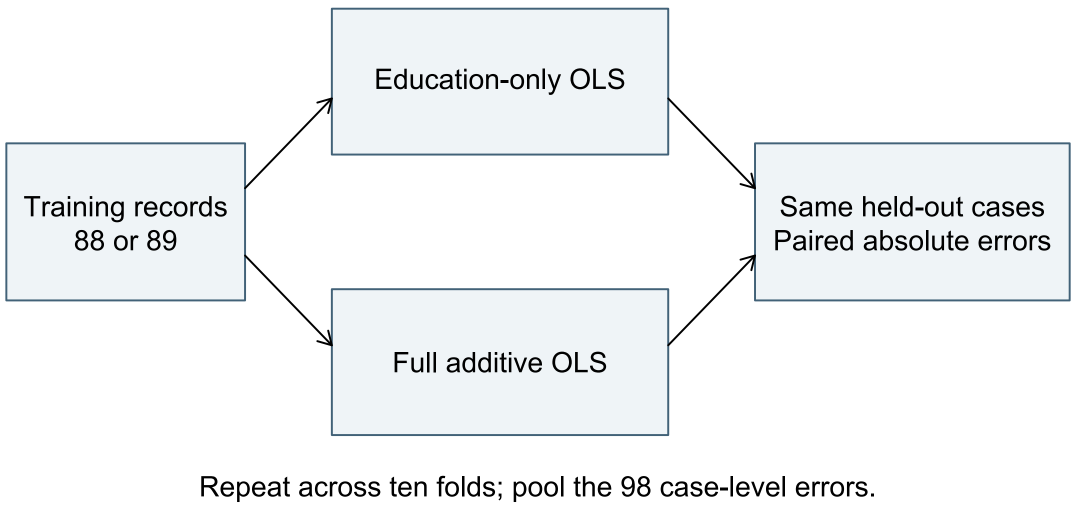

Figure 5: Shared held-out cases let two fixed learning rules face the same prediction task.

6.3 Work one error, then read the comparison

For government administrators, the first source record, observed income is 12.351 thousand dollars. Its held-out fold is 9. The full-rule prediction is 11.594819, giving

$$|12.351 - 11.594819| = 0.756181.$$

The education-only prediction is 8.946130, giving $|12.351 - 8.946130| = 3.404870$. This one record favors the full rule. We still need all 98 errors:

Table 8

Case-weighted absolute errors from the frozen shared-fold comparison.

Learning rule	Held-out MAE (thousand dollars)
Education-only OLS	2.403
Full additive OLS	1.541
Difference: full minus education-only	-0.862

The full rule's average held-out absolute error is about 862 dollars lower in this split. Because we subtract the two errors for **each same case** before averaging, difficulty differences between validation cases do not come from using different test sets for the two rules.

What is the performance estimand? Eight test folds contain 10 cases whose predictions were trained on 88; two contain 9 cases trained on 89. Consequently the pooled quantity targets the declared mixture

$$R_{\text{mix}} = \frac{80}{98} R_{88} + \frac{18}{98} R_{89},$$

where R_m is expected absolute error of the specified learning rule trained on m eligible same-source records and used on a new one under the stated repetition. Pair 3 compares the two rules' R_{mix}. It does not directly estimate performance of the single equation fitted to all 98 observed records.

The 200 resplits give full-minus-education MAE differences from -0.914 to -0.741, all favoring the full rule. This supports stability to the examined split choices. It is **not a 95% interval**: the same cases are reused, training sets overlap, and the results are not independent studies. We do not test a mean of ten fold errors as if those errors were independent. Uncertainty across newly drawn training samples is not quantified here. The estimated risk advantage is therefore provisional; its sign is not established for the hypothetical source population by the resplit range.

The comparison supports lower internal held-out error for the full rule in these records and under this protocol. It does not isolate the predictive contribution of education: the rules differ by prestige, women, and type together. Nor does it establish external transport, individual earnings prediction, or a causal education effect.

7 Read the Coordinated Arguments, Then Evaluate and Synthesize

7.1 The chapter-authored study record

The following report consolidates the preceding analysis. Treat its paragraphs, tables, equations, and code as an accessible teaching record, not as statements copied from the source investigators.

> **R0 – Common record and declared scope.** We analyzed the 98 complete occupation records described in Section 1. Income was scaled to thousand 1971 Canadian dollars. Education and income were historical occupational averages; prestige came from an earlier rating survey. We retained the same rows for all three questions and make no claim of verified probability sampling or randomized education.
>
> **R1 – Marginal answer.** Across these records the education-only least-squares slope was 0.883 thousand dollars per year. This is a descriptive linear summary: occupations with more education tended to have higher average income. It is not an individual or causal effect.
>
> **R2 – Conditional answer.** The intended estimand was the linear conditional-mean education coefficient β_E, comparing education at the same prestige, percentage of women, and occupational type. Its fitted estimate was $b_E = 0.116$, with a classical 95% interval [-0.430, 0.663] under the independent linear conditional-mean working model with common variance. HC3 gave [-0.643, 0.876]. Influence, functional-form, and support checks limit that model's interpretation. The direction and practically relevant magnitude remain unresolved under these calculations; the result does not establish no conditional relationship.
>
> **R3 – Prediction answer.** The two fixed OLS learning rules were compared with the shared ten-fold protocol in Section 6. The full rule had MAE 1.541 versus 2.403 for education-only, difference -0.862. The examined resplits all favored the full rule. This is internal evidence about learning procedures trained on 88/89 eligible records under the stated same-source exchangeability assumptions. Lower expected error is a provisional conclusion because new-sample uncertainty is not quantified; this is not external validation of a final 98-record equation.
>
> **R4 – Synthesis.** Education has a positive marginal linear association with income in these historical records. Its conditional association is imprecisely resolved by the specified additive model, while the full learning rule predicts the held-out records more accurately than the education-only rule. These conclusions answer different questions and do not establish that changing education would change income.

7.2 Read the functions and their traces

A component is a logical function; the record contains study-specific content serving that function. A passage can carry several functions. For example, R0 identifies records for STAT-E and supplies historical context for SUB-E. It becomes J content only when it offers a reason connecting those records to a particular inference. "We used R" does not by itself supply that reason.

Table 9

Reading the linked-layer functions without three duplicate maps.

Function	Shared or pair-specific content	Trace in this teaching record
STAT-E	Named 98 rows, variables, exclusions, and computations' inputs	R0; the row audit and import code in Section 1
SUB-E	Occupations rather than workers; historical dollars, education, ratings, and types	R0; variable/source descriptions
STAT-J	Pair 1: fixed-record least-squares definition. Pair 2: declared stochastic model and uncertainty conditions. Pair 3: held-out fitting, fixed rules, loss, and exchangeability	R1–R3; model declaration; validation protocol and code
SUB-J	Reasons connecting those measurements and comparisons to historical occupational income, with limits on individual, current, and causal interpretations	R0–R3; unit/time explanations and prediction-opportunity definition
STAT-I	Pair 1's .883 slope; Pair 2's .116 estimate, intervals, coefficient test and checks; Pair 3's case-level errors, MAEs, and split sensitivity	Section 3 through Section 6; R1–R3
STAT-C	Descriptive positive marginal slope; unresolved conditional slope under the model; smaller internal error for the full rule under the protocol	R1, R2, and R3 respectively
SUB-I	Interpret years, income units, conditioning, and prediction errors in the named historical occupation setting	Worked substitutions and error calculation; R1–R3
SUB-C	Each question-facing answer with its own limits; none supplies a causal or individual-earnings answer	R1–R3; their distinctions preserved in R4

SUB-E situates evidence; SUB-J **augments and contextualizes** statistical justification; SUB-I interprets the inference; SUB-C bounds the substantive answer. Sharing a dataset does not make Pair 2's independent-error model into Pair 3's validation procedure.

The four documentary statuses from Chapter 1 remain useful. Here, **stated in the paper** means stated in the chapter-authored report; **located in related materials** covers the linked source documentation and executable calculation record; **implicit but traceable** marks an identifiable link whose premise is actually in the record; and **not located** records a missing trace in the accessible material. A separate conflict flag records inconsistent sources.

For example, R2 states the independence assumption. Evidence establishing that assumption for the original occupation-selection process is **not located** in this teaching record. Those are different entries: a stated assumption is not a demonstrated sampling design. Conversely, the translation of .883 thousand dollars into about 883 dollars is implicit but traceable through the stated unit conversion; no new empirical premise is invented. No conflict is identified in the defined teaching record; a conflict search in a real study could have a different result.

Professional knowledge helps identify what function a trace serves. It does not authorize adding an unreported design feature. If an author gave the invalid reason "all rows are in the same dataset, therefore they are independent," it would still be **stated J**, whose adequacy is judged separately. Reading records the claim actually offered, not a repaired claim the reader wishes had been offered.

7.3 Evaluate each argument separately

We now change from identifying the reported reasoning to evaluating its quality. The six Chapter 1 questions apply to each argument. They are not six scores to total, and not every concern needs to be counted repeatedly.

Table 10

Apply the six questions before evaluating the study-level synthesis.

Quality-evaluation question	Application to these arguments
Question alignment	R1 matches its finite-record slope. R2 names β_E, its fitted estimate, the conditioning set, and the model extension. R3 names the learning procedure, loss, and training sizes rather than substituting a coefficient test.
Evidence integrity	The 102-to-98 accounting and common rows are explicit. Missing type still limits what is represented. CV excludes held-out outcomes from fitting; source selection is not repaired by that procedure.

Quality-evaluation question	Application to these arguments
Justification adequacy	Least squares is sufficient for the declared finite description. Pair 2's model conditions and Pair 3's same-source exchangeability remain assumptions with incomplete empirical support. Statistical computation alone cannot establish them.
Inferential adequacy	The slope, intervals, error arithmetic, and comparisons are reproducible. Neither zero-crossing nor a large p-value proves absence; repeated folds are not independent replication. Substantive reasoning must retain those distinctions.
Transition fidelity	Occupation-level average income remains occupation-level average income; partial adjustment is not a causal intervention, and model fit is not held-out performance.
Claim calibration	R1 remains finite and historical; R2 reports unresolved magnitude/direction rather than no relationship; R3 remains internal and procedure-specific. No claim extends to individual workers or modern populations.

At an individual question, useful judgments include **adequate for this purpose**, **needs revision**, **insufficient information to judge**, or **not applicable**, with a reason. For example, the record is adequate to check Pair 3's held-out computations but gives insufficient information to establish exchangeability for an external population. These judgments are not documentary statuses.

Each whole pair then receives its own conclusion:

- **Pair 1: supported as stated.** The answer is an accurate bounded description of the retained records.
- **Pair 2: supported as stated**, as the explicitly conditional, unresolved working-model answer about β_E actually reported. This does not certify the model or establish a population relationship. A claim that the true national conditional slope is zero would not be supported by this record.
- **Pair 3: supported as stated**, as a qualified internal comparison with assumptions and limits retained. External superiority of the final all-record fitted equation is **not assessable from the accessible record**; it is not the reported answer being accepted.

The other available whole-pair conclusions are **supported only after narrowing or revision**, **not supported**, and **not assessable from the accessible record**. A correctly reported inconclusive answer can be supported as stated. Evaluating what the study actually says prevents a bounded answer from being rejected merely because it does not answer a stronger question.

7.4 Add Cross-question coherence for the synthesis

For coordinated questions, ask one additional question: **Does the synthesis preserve the distinct questions, estimands, methods, analytic statuses, uncertainty, and limits of the separate answers?** This is **Cross-question coherence**, introduced locally here. Chapter 1a develops it further but is not needed to apply it.

R4 is coherent because it reports three distinct outcomes. A positive marginal slope, an unresolved conditional slope, and better internal prediction can coexist. They do not contradict one another.

Now consider the alternative sentence:

> Once the other predictors were controlled, education had no effect; nevertheless, the significant full regression proves it will predict workers' incomes accurately.

Reading first identifies the offered inference and claim. Evaluation then finds several failures: "no effect" substitutes a causal null for the conditional-association answer; an omnibus test is not validation; and occupations become workers. No omnibus significance test can replace the held-out errors or rescue the change of unit. The corrected synthesis is R4, not a fourth overarching argument that erases the separate verdicts.

7.5 Auxiliary construction callback

The construction workflow supports researchers carrying out the study; it is not a mandatory reading sequence. Each question retains its own six-stage cycle from Chapter 1. For coordinated questions, Chapter 1a's extension aligns the plans at Stage 3 and adds Stage 7 for synthesis.

In **Understand and plan**, researchers review the problem and literature, frame questions and candidate answers, and align design, measurement, estimands, evidence acquisition, and inference. Here, "separate questions" means separately owed conclusions. If the obligation were one integrated conclusion, jointly necessary estimands would be planned as strands instead.

The extended **Stage 3, Plan and align prospectively across questions**, specifies common rows and units but distinct marginal, model-based, and validation plans. In **Produce and complete**, **Stage 4, Produce or acquire and document evidence**, designates the question-specific records; **Stage 5, Estimate, check, and analyse**, performs the computations; and **Stage 6, Contextualize, justify, interpret, and complete each bounded substantive answer**, finishes each answer with its actual limits. **Stage 7, Give explicit outcomes and synthesize**, records all outcomes and brings them together without creating an omnibus argument.

Iteration remains within a phase. A same-anchor error correction or justified uncertainty repair can reopen Phase 2 work. A newly preferred outcome, question mode, or estimand requires a new or revised cycle, with reused records where appropriate and its timing disclosed. A newly chosen causal question cannot be "narrowed" into association and treated as answered. This teaching reanalysis does not retrospectively claim that the historical data were collected for our three questions.

The occupation study supplies no randomized-design warrant. A second application now shows how covariate adjustment can seek the same assignment-effect estimand when design and weighting support that link.

8 Project STAR: A Randomized-Design Transfer

This is the full STAR transfer signposted from Chapter 7. It first connects binary regression to observed arm means, then compares adjusted and unadjusted specifications on identical primary rows. These are related calculations, not interchangeable estimates from one unchanged sample.

8.1 A different design supplies a different warrant

Project STAR supplies a transfer with its own question:

> Among eligible kindergarten pupils in the participating Tennessee schools, what was the mean reading-score difference under assignment to a small rather than a regular class?

The outcome is *readk*, an end-of-kindergarten reading scale score. The variable *stark* records the class-type condition. Interpreting its small-versus-regular comparison as an assignment comparison rests on the documented experimental protocol, not on the column name or an *lm()* call. Random assignment occurred within participating schools (Finn & Achilles, 1990; Mosteller, 1995). The regular-plus-aide condition is outside this declared two-arm comparison. Class type is not a measured number of pupils, and this analysis does not estimate a headcount dose response.

The package field does not independently verify each pupil's original randomized assignment. An assignment-effect reading additionally requires linking that field to the assignment record and checking switches or implementation deviations. This teaching reanalysis does not supply that pupil-level verification.

The data are distributed as *AER::STAR*, accompanying *Applied Econometrics with R* and following the version used by Stock and Watson (Kleiber & Zeileis, 2008; Stock & Watson, 2007). They are loaded from the package rather than redistributed here. Read the package's variable-format list: the chosen outcome is reading, not an invented combined reading/math score. The teaching analysis is retrospective, not preregistered.

The scientific target is the pupil-weighted mean assignment effect for the eligible kindergarten cohort in participating schools. A complete-record common school-fixed-effect coefficient is an estimator requiring additional links to that target. Randomization does not, by itself, guarantee those links after exclusions or under heterogeneous effects.

8.2 Separate eligibility from missingness

Some distributed pupil records belong to later entrants and have no kindergarten condition recorded. Their structural absence is distinct from missing reading or covariates among eligible kindergarten pupils. Only variables required for this analysis enter its complete-record rule; *mathk* must not exclude an otherwise usable reading record.

```
data("STAR", package = "AER")
star_vars <- c("stark", "readk", "lunchk", "gender",
               "ethnicity", "experiencek", "schoolidk")
star_k_recorded <- !is.na(STAR$stark)
star_later_recorded <- rowSums(!is.na(
  STAR[, c("star1", "star2", "star3")]
))
star_two_arm <- star_k_recorded &
  STAR$stark %in% c("regular", "small")
star_complete <- complete.cases(STAR[, star_vars])
star_df <- droplevels(STAR[star_two_arm & star_complete, star_vars])
star_df$stark <- relevel(star_df$stark, ref = "regular")
star_observed <- droplevels(STAR[
  star_two_arm & !is.na(STAR$readk),
  c("stark", "readk", "schoolidk")
])
star_observed$stark <- relevel(star_observed$stark, ref = "regular")
c(observed_reading = nrow(star_observed),
  complete_covariates = nrow(star_df))
```

```
   observed_reading complete_covariates
               3745                3735
```

Table 11
STAR eligibility and reading-analysis completeness.

Stage	n
Distributed records	11598
Kindergarten condition recorded	6325
Eligible small or regular	4094
Eligible, reading observed	3745
Complete primary records	3735
Primary regular	2001
Primary small	1734
Primary schools	79

Of 11,598 distributed records, 5,273 have no kindergarten condition; all have later-grade participation in this file. They are structural late entrants, not 5,273 failed kindergarten reading measurements. The remaining 6,325 have a recorded kindergarten arm. Across all three arms, 536 lack reading, leaving 5,789 observed-reading records. These are counts in the distributed file, not a replacement for every original trial enrollment figure.

Among the 4,094 eligible small/regular records, 349 lack reading. The 3,745 observed-reading records contain 2,006 regular-arm and 1,739 small-arm pupils; the corresponding missing-reading counts are 188 and 161. Similar missing percentages do not establish ignorable missingness: the absent outcomes could differ systematically within either arm.

Do not discard records for variables unused by a calculation. Across all 5,789 observed-reading records, 38 lack at least one of lunch status, gender, ethnicity, or teacher experience. Within the focal two arms, ten records lack a required adjustment covariate. The primary common-covariate sample therefore has 3,735 pupils: 2,001 regular and 1,734 small, in 79 schools. The observed-reading sensitivity retains all 3,745 focal records; no further observed-reading records lack school ID.

```r
star_covariates <- c("lunchk", "gender", "ethnicity", "experiencek")
star_reading_recorded <- star_k_recorded & !is.na(STAR$readk)
c(missing_covariates = sum(star_reading_recorded &
    !complete.cases(STAR[, star_covariates])),
  regular_missing_reading = sum(STAR$stark == "regular" &
    is.na(STAR$readk), na.rm = TRUE),
  small_missing_reading = sum(STAR$stark == "small" &
    is.na(STAR$readk), na.rm = TRUE))
```

```
     missing_covariates regular_missing_reading 
                     38                     188 
  small_missing_reading 
                    161 
```

Complete-record inference requires an assumption that selection does not distort the relevant within-school arm comparison, given the declared covariates and target. The observed-reading sensitivity relaxes the additional covariate-completeness restriction; it does not recover the reading scores of the other 349 eligible pupils.

For a full-cohort causal contrast, observation must be connected to potential reading scores. One possible assumption is that, within arm and school, observation does not depend on those scores, with adequate observation in both arms and appropriate school weighting. Observed outcomes alone cannot verify that assumption. Selecting observed outcomes can otherwise destroy the comparability created by randomization.

8.3 The binary slope recovers the observed arm means

Let $D_i = 1$ for the recorded small arm and 0 for regular. In the model $\widehat{Y_i} = \hat{\alpha} + \hat{\delta} D_i$, minimizing RSS fits each arm's observed mean separately. Thus

$$\hat{\alpha} = \bar{Y}_{D=0}, \qquad \hat{\delta} = \bar{Y}_{D=1} - \bar{Y}_{D=0}.$$

```r
star_observed_unadjusted <- lm(readk ~ stark, data = star_observed)
star_arm_means <- tapply(star_observed$readk, star_observed$stark, mean)
star_observed_difference <- unname(
  star_arm_means["small"] - star_arm_means["regular"]
)
c(regular_mean = unname(star_arm_means["regular"]),
  small_mean = unname(star_arm_means["small"]),
  small_minus_regular = star_observed_difference,
  regression_difference = unname(
    coef(star_observed_unadjusted)["starksmall"]
  ))
```

```
         regular_mean            small_mean   small_minus_regular
              434.732               440.547                 5.815
regression_difference
                5.815
```

On these 3,745 records, $440.547 - 434.732 = 5.815$ reading points. The equality is algebraic and does not require random assignment. Under independent common-variance rows, the ordinary slope test would equal the pooled two-sample t test; that row-level reference does not address this design's within-school dependence.

The raw arm means weight schools by each arm's observed pupil counts, and those weights can differ by arm. Common assignment probabilities and suitable observation patterns, or explicit standardization to target school weights, are needed to link the pooled difference to the pupil-weighted cohort target. Clustering the SE does not adjust the coefficient for school blocks, missing outcomes, or unequal target weights.

8.4 Compare two specifications on identical primary rows

The unadjusted model reproduces the small-minus-regular mean difference on the primary records. The adjusted model adds free-lunch status, gender, ethnicity, teacher experience, and school indicators. School indicators respect the assignment blocks; the other recorded covariates are the declared adjustment specification, not a substitute for randomization.

```r
star_unadjusted <- lm(readk ~ stark, data = star_df)
star_primary <- lm(
  readk ~ stark + lunchk + gender + ethnicity +
    experiencek + schoolidk,
  data = star_df
)
star_sensitivity <- lm(
  readk ~ stark + schoolidk, data = star_observed
)
c(unadjusted = nobs(star_unadjusted),
  adjusted = nobs(star_primary),
  observed_reading = nobs(star_sensitivity))
```

```
      unadjusted         adjusted observed_reading 
            3735             3735             3745 
```

Adjustment can improve precision when added variables are prognostic. It is not necessary to remove baseline confounding created by a successfully implemented random assignment. It also does not automatically preserve every causal target: variable timing, selection, and the fitted comparison still need scrutiny.

On the common 3,735 primary rows, the unadjusted difference is 5.895465 and the adjusted coefficient is 6.657510. The earlier 5.815 difference uses 3,745 observed-reading rows; it must not serve as the supposedly same-row unadjusted baseline. The observed-reading school-adjusted sensitivity is another specified calculation, not the raw observed mean difference.

The common *starksmall* coefficient is a coefficient of the stated additive projection. If effects vary across schools or profiles, its implicit weights need not equal the pupil weights defining the cohort target. A common effect or a target-matching weighting argument is therefore an additional requirement. Similar unadjusted and adjusted coefficients are consistent with successful randomization, not proof that implementation, missingness, or weighting problems are absent.

8.5 Cluster uncertainty follows the design

Pupils are outcome units; schools are the approximately independent units for this covariance calculation and the randomization blocks. Schools themselves were not assigned to the two arms. Pupils sharing a classroom may also be dependent; school clustering allows within-school residual association, including that within classrooms, under an independent- or weakly-dependent-school large-cluster approximation.

For school g, form its score $s_g = X_g' e_g$. The CR1/HC1 covariance used here combines residual contributions **within school before taking their outer product**:

$$\widehat{V}_{\text{CR1}} = c_G (X'X)^{-1} \left(\sum_{g=1}^{G} s_g s_g' \right) (X'X)^{-1}, \qquad c_G = \frac{G}{G-1} \frac{n-1}{n-p}.$$

Here p counts all fitted coefficient columns, including the intercept; for the binary-only model $p = 2$. The within-school cross-products distinguish this from treating every pupil as independent. The implementation below uses school-clustered HC1 covariance, the finite-cluster correction, and a t_{G-1} critical value. These are stated approximations, not exact randomization inference.

```
star_V <- sandwich::vcovCL(
  star_primary, cluster = star_df$schoolidk,
  type = "HC1", cadjust = TRUE
)
star_b <- unname(coef(star_primary)["starksmall"])
star_se <- sqrt(star_V["starksmall", "starksmall"])
star_G <- nlevels(star_df$schoolidk)
star_critical <- qt(.975, star_G - 1)
c(estimate = star_b, SE = star_se)
```

```
estimate       SE
   6.658    1.697
```

```
star_b + c(-1, 1) * star_critical * star_se
```

```
[1]  3.278 10.037
```

This example uses the primary adjusted model and the school IDs from its 3,735 records. Apply the same calculation to the other fitted models using the school IDs from each model's own rows. The table collects their results; changing the model or its records is distinct from changing the covariance calculation.

Table 12
Class-type comparisons and clustered intervals.

Model	n	Estimate	SE	Interval
Observed: unadjusted	3745	5.815	1.848	[2.14, 9.49]
Primary: unadjusted	3735	5.895	1.849	[2.21, 9.58]
Primary: adjusted	3735	6.658	1.697	[3.28, 10.04]
Observed: school-adjusted	3745	6.627	1.780	[3.08, 10.17]

For the unadjusted 3,745-record comparison, the 79-school covariance gives SE 1.847617. With $t_{.975,78} \approx 1.991$, its approximate interval is $[2.137, 9.493]$ around 5.815. This preserves the binary-mean comparison while changing its uncertainty calculation; it does not repair missing outcomes or school weighting.

The primary adjusted estimate is 6.658 reading points, with approximate interval $[3.278, 10.037]$. The observed-reading sensitivity is 6.627, with interval $[3.084, 10.170]$. The similarity is informative about this limited sensitivity; it is not evidence that all missingness is ignorable.

These intervals rely on independent schools and a cluster-asymptotic approximation with 79 clusters. They are neither exact randomization intervals nor results from a multilevel model. Row-level HC3 would address unequal variance but not within-school dependence. A residual plot cannot choose the design's independence unit. Nor does a count of 79 schools certify the approximation: information balance, dependence, and the influence of particular clusters also matter.

8.6 Read the bounded answer, then evaluate its support

A chapter-authored report can state:

> On 3,735 complete two-arm kindergarten records, the common adjusted small-versus-regular class-type coefficient was 6.658 reading points, with approximate school-clustered interval [3.278, 10.037]. Its causal mean-effect interpretation relies on documented within-school randomization, a verified link from recorded class type to original assignment, suitable complete-record selection, independent-school uncertainty, and a common-effect or target-matching weighting argument. It concerns assigned class type, one kindergarten reading outcome, and participating Tennessee schools, not realized headcount or a portable policy dose response.

The code and sample audit carry **STAT-E** content; the design and covariance paragraph offer **STAT-J** reasons; the estimates and interval calculation supply **STAT-I** content; and the bounded statistical answer supplies **STAT-C** content. **SUB-E** situates the pupils, test, cohort, and school context. **SUB-J** augments and contextualizes the justification through the assignment record and the selection/weighting qualifications. **SUB-I** interprets the comparison as assignment-related only under that support. **SUB-C** bounds the substantive answer. These are linked logical functions within one argument; neither layer is a separate argument.

The chapter-authored report states the selection and weighting conditions; stating them does not establish that they hold. The randomization protocol is located in related study publications, not demonstrated by the coefficient table. Reading first locates these traces and their documentary statuses. A separate evaluation asks whether they adequately support the target and stated conclusion. If a necessary link cannot be assessed, report that limitation rather than inventing it or silently replacing the target with a different one.

Unlike Prestige, this example has an experimental assignment warrant. Neither a stable coefficient nor better prediction would have created that warrant. Under adjustment, a focal coefficient moves through the added predictors' relations with the focal predictor and outcome, conditional on terms already included. Within-school randomization makes baseline relations with assignment small on average; prognostic relations with reading may remain. This explains why useful adjustment can retain an assignment target without treating every model change as a new scientific question.

The primary design publications and subsequent analyses document implementation and interpretation issues (Chetty et al., 2011; Finn & Achilles, 1990; Hanushek, 1999; Krueger, 1999; Mosteller, 1995; Nye et al., 2000). The present result does not settle transport to nonparticipating schools, another period, another outcome, or another education system. Those limits belong in the answer, not only in a list after an unrestricted causal claim.

8.7 Apply the randomized-design transfer

Reproduce the eligibility, observed-reading, complete-record, arm, and school counts. Explain why *mathk* does not enter the exclusion rule. Recover the binary slope from the two observed arm means and reproduce its school-clustered interval. Fit the two primary models on identical rows, then the observed-reading school-adjusted sensitivity. Keep their samples and coefficients distinct. Identify the documented reasons for school blocking and clustered uncertainty, and distinguish the fitted coefficient from the pupil-weighted cohort target. Read the report's STAT/SUB functions and trace statuses before separately evaluating its selection, weighting, causal, outcome, and transport boundaries.

9 Summary and Reading Bridge

Multiple regression adds conditional comparisons, not automatic causal control. Subtracting fitted means shows what a coefficient compares; residualization connects it to covariance. Residual, influence, support, and precision checks have different consequences for a bounded answer. A fitted profile and a standardized fitted quantity are distinct didactic extensions of the conditional model; held-out predictive performance answers a separately owed question. Changing rows, conditioning, coding, or uncertainty must be identified by what actually changes, not treated as interchangeable operations.

The occupation study owes three answers. Its marginal association is a finite-record description. Its conditional interval is a working-model inference with limited stability and unresolved magnitude. Its held-out errors address a separate learning-rule performance question. Shared records make the questions coordinated; they do not make their justifications identical. Read each answer before evaluating quality and then checking the coherence of the synthesis.

The STAR transfer changes the warrant, not the need for careful alignment. Documented within-school assignment can support an assignment-effect reading, while missing outcomes, recorded assignment, school weighting, common-row adjustment, and cluster-based uncertainty still limit the answer. The 3,745-row observed mean difference and the 3,735-row adjusted coefficient are useful because their samples and functions remain explicit, not because their similar values prove every link.

For further study:

- The OpenIntro multiple-predictor tutorial offers concrete equations and indicator interpretations. We use that equation-first teaching device without adopting automatic stepwise selection as this chapter's route.
- The ISLR regression lab connects fitting, coefficients, checks, and predictions.
- The ISLR cross-validation lab develops the train/predict/check sequence. Our fixed rules, MAE, and training-size qualifications are specified for this teaching example.
- The carData documentation supplies the Prestige variable meanings and provenance. The coordinated study and R0–R4 report are chapter-authored, not claims attributed to that documentation.

Chapter 1a offers an optional extension of argument coordination. The next regression question is different: might the education comparison itself vary with another predictor, rather than remain the same additive slope? That is the reason to introduce interaction.

Exercises

Use the supplied numbers; rounding within 0.02 thousand dollars is acceptable unless a different precision is requested. These prompts do not require Chapter 1a or the appendices.

Exercise 1: Keep the three answers distinct (Foundational)

A report promises: (a) a description of education and income across 98 occupations, (b) an education comparison at the same prestige, women, and type, and (c) a comparison of two prediction rules on held-out records.

1. Explain why these are three question–answer pairs rather than three statistical strands serving one integrated answer.
2. For each, name its estimand in plain language and one kind of output that could contribute to its answer.
3. Does sharing the same 98 records establish the same justification for all three? Give one concrete reason.

Exercise 2: Name what changes (Foundational)

Classify each change as primarily changing the records, the comparison, the coefficient coordinates, or the uncertainty calculation. State whether it necessarily creates a new scientific question.

1. Refit education-only regression after excluding four missing-type records.
2. Add prestige, women, and type to the education-only association equation.
3. Change the reference category from blue-collar to white-collar.
4. Keep the fitted equation but replace classical SEs with HC3 SEs.

Finally, explain why appropriately specified adjustment in a randomized study need not change its declared average assignment-effect estimand.

Exercise 3: Interpret and calculate a fitted profile (Foundational)

Use this rounded hypothetical equation, with income in thousand dollars:

$$\hat{y} = 1 + 0.20e + 0.10p - 0.04w + 0.30D_{\text{prof}} + 0.15D_{\text{wc}}.$$

1. Calculate the fitted mean for a white-collar profile with $e = 12$, $p = 40$, and $w = 50$.
2. Calculate its modeled difference from a profile with $e = 10$ and the same other values. Interpret the result in dollars.
3. What must be checked before calling this a comparison supported by the observed data?
4. Why is neither calculation an estimate of the effect of giving a worker two more years of education?

Exercise 4: Read held-out errors (Foundational)

For four held-out occupations, observed incomes are $(5, 8, 10, 13)$ thousand dollars. The education-only predictions are $(4, 10, 7, 12)$; full-rule predictions are $(6, 8, 9, 11)$. Each prediction was fitted without the corresponding outcome.

1. Calculate the four absolute errors and MAE for each rule.
2. Calculate the average paired difference, full minus education-only.
3. Does the better average rule have to be better for every occupation?
4. Explain why this comparison is neither a coefficient interval nor validation of a model subsequently fitted to all available records.

Exercise 5: Locate a reason before judging it (Integrative)

An accessible report states:

> We retained the same 98 occupations for both equations. This makes the observations independent. The full equation has higher training R^2, so we conclude it predicts new occupations more accurately.

The report provides import code and fitted-model output, but no validation split, held-out predictions, or study-specific evidence about dependence.

1. Identify the reasons offered, the inference, and the claim.
2. Record the documentary status of the independence reason and the held-out prediction evidence separately. Do not rewrite "stated" as "not located" merely because a reason is wrong.
3. Apply Justification adequacy, Inferential adequacy, and Claim calibration.
4. Give a whole-pair evaluation conclusion for the prediction answer and name the additional work it would require.

Exercise 6: Evaluate the study-level synthesis (Integrative)

Use the occupation study's reported results: marginal slope .883; fitted conditional estimate $b_E = .116$ of β_E, with classical interval [-.430, .663]; and held-out MAEs 2.403 for education-only versus 1.541 for the full rule.

A summary says:

> Education does not matter after adjustment, but the model proves that increasing workers' education raises income and predicts earnings well in other countries.

1. Read the claims actually offered before evaluating them.
2. Identify which separate answer obligation is being replaced or exceeded by each clause.

3. Apply Cross-question coherence and write a corrected three-sentence synthesis that retains the three reported answers and their limits.
4. If the authors now want a causal effect of education, is that a repair of their existing uncertainty calculation or a new/revised question-level cycle? Explain.

Appendix A: Adjustment, Coding, and Group-Comparison Mathematics

This appendix develops the mathematics behind Section 2. Appendix B retains technical reproduction and sensitivity; Section 8 changes the design context. The occupation calculations use the same 98-row object *d* and the models *m0*, *m1*, and *m2* defined in the main text. They do not introduce another sample. The type codes remain *bc*, *prof*, and *wc*.

Why the marginal slope includes another association

Suppose a population process is

$$Y = \beta_0 + \beta_X X + \beta_Z Z + \varepsilon, \qquad E(\varepsilon \mid X, Z) = 0,$$

with finite variances and $\mathrm{Var}(X) > 0$. The marginal least-squares slope is a covariance divided by a variance. Substituting the outcome equation gives

$$\begin{aligned} \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)} &= \frac{\beta_X \mathrm{Var}(X) + \beta_Z \mathrm{Cov}(X,Z) + \mathrm{Cov}(X,\varepsilon)}{\mathrm{Var}(X)} \\ &= \beta_X + \beta_Z \frac{\mathrm{Cov}(X,Z)}{\mathrm{Var}(X)}. \end{aligned} \tag{1}$$

The zero-error-covariance condition removes the final term. The discrepancy therefore involves both the association between X and Z and Z's coefficient in the outcome model. Neither ingredient alone determines its size. If the target is the conditional coefficient, omission creates bias relative to that target. If the target is the marginal slope, that marginal quantity is not a biased version of an unstated conditional target. Neither interpretation supplies a causal warrant for observational variables.

Changing a specification does not universally change a scientific estimand. Different estimators can target the same declared quantity, as the randomized adjustment example in Section 8 illustrates. Releveling a factor merely changes coordinates. By contrast, the marginal and conditional coefficients in this example name different comparisons.

A known process and repeated samples

Let independent standard normal variables X and U generate $Z = .65X + \sqrt{1 - .65^2}U$. Then both predictors have variance one and their correlation is .65. Independently generate a normal error with SD 6, and set

$$Y = 50 + 2X + 5Z + \varepsilon. \tag{2}$$

The marginal slope is $2 + 5(.65) = 5.25$; the conditional X slope is 2. The following single sample has 250 observations. Its measured strategy and prior-attainment labels are fictional teaching interpretations.

```
RNGkind("Mersenne-Twister", "Inversion", "Rejection")
set.seed(82108)
ch8a_n <- 250
ch8a_x <- rnorm(ch8a_n)
ch8a_z <- .65 * ch8a_x + sqrt(1 - .65^2) * rnorm(ch8a_n)
ch8a_y <- 50 + 2 * ch8a_x + 5 * ch8a_z +
  rnorm(ch8a_n, sd = 6)
ch8a_sim <- data.frame(
  strategy = ch8a_x, prior = ch8a_z, score = ch8a_y
)
ch8a_marginal <- lm(score ~ strategy, data = ch8a_sim)
ch8a_adjusted <- lm(score ~ strategy + prior, data = ch8a_sim)
c(marginal = unname(coef(ch8a_marginal)["strategy"]),
  conditional = unname(coef(ch8a_adjusted)["strategy"]))
```

```
   marginal conditional
      5.997       3.026
```

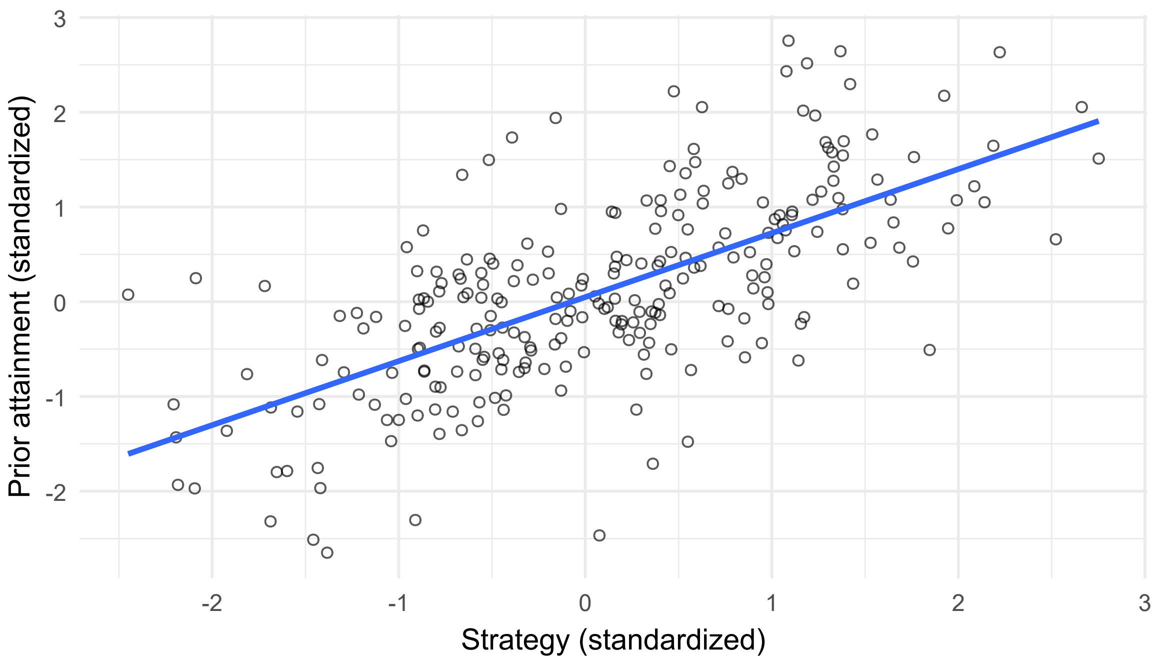

Figure 6: Joint predictor support in one generated sample.

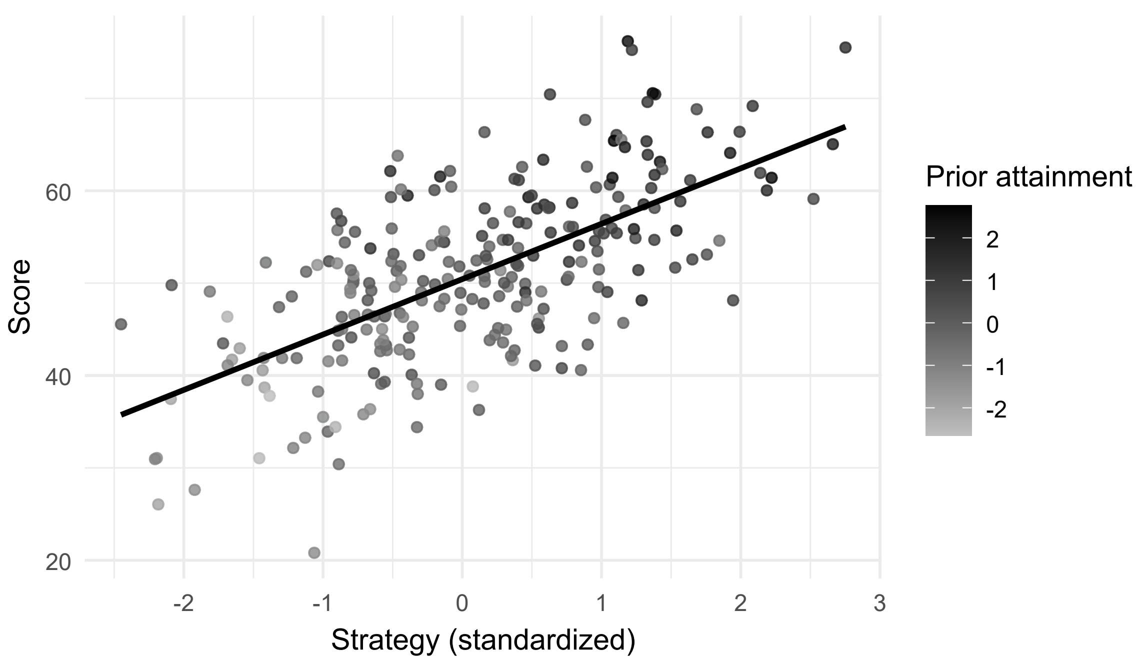

Figure 7: A marginal line does not hold prior attainment fixed.

The first plot displays the support needed for a conditional comparison. The second shows that prior attainment varies along the marginal relation. A line calculated outside the observed predictor cloud is not made well supported by the existence of its equation.

Now use a different, explicitly stated sample size: 150 observations in each of 1,000 repetitions. This retains the earlier repeated-sample exercise, rather than silently treating it as repetitions of the 250-observation draw.

```
set.seed(82109)
ch8a_one_rep <- function() {
  x <- rnorm(150)
  z <- .65 * x + sqrt(1 - .65^2) * rnorm(150)
  y <- 50 + 2 * x + 5 * z + rnorm(150, sd = 6)
  c(Marginal = unname(coef(lm(y ~ x))["x"]),
    Conditional = unname(coef(lm(y ~ x + z))["x"]))
}
ch8a_slopes <- t(replicate(1000, ch8a_one_rep()))
```

Table 13

Targets and results across 1,000 generated samples.

Model	Target	Mean	SD
Marginal	5.25	5.258	0.595
Conditional	2.00	2.001	0.671

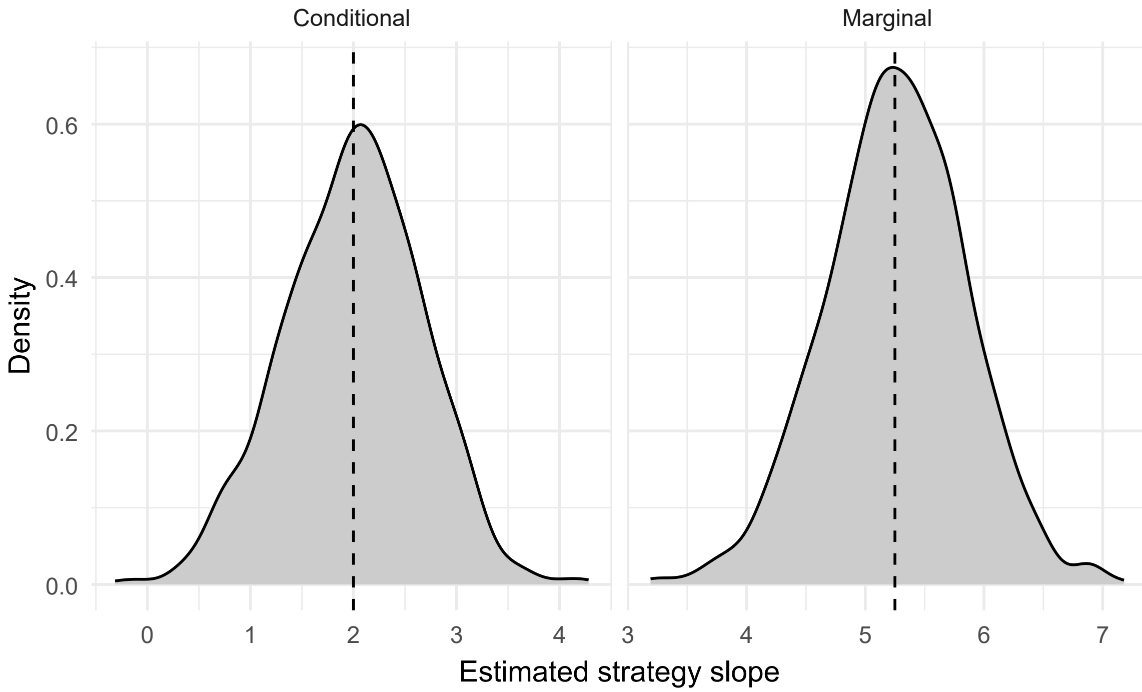

Figure 8: Sampling distributions around two different targets.

Both procedures center near their own targets. The conditional estimates are more variable here: separating correlated predictors uses less residual predictor variation. This is a result of the specified process, not a general promise that adjustment always lowers precision. Prognostic covariates and their relations to the focal predictor both matter. Simulation demonstrates consequences of its assumptions; it cannot certify the observational occupation study.

Indicators recover fitted group means

Start with a model containing only the three-level type factor. With blue-collar as reference,

$$\hat{y}_g = \hat{\alpha} + \hat{\delta}_{\text{prof}} D_{\text{prof}} + \hat{\delta}_{\text{wc}} D_{\text{wc}}.$$

The fitted means are $\hat{\mu}_{\text{bc}} = \hat{\alpha}$, $\hat{\mu}_{\text{prof}} = \hat{\alpha} + \hat{\delta}_{\text{prof}}$, and $\hat{\mu}_{\text{wc}} = \hat{\alpha} + \hat{\delta}_{\text{wc}}$. OLS minimizes squared deviations separately within each group, so these equal the observed group means. The hats distinguish fitted summaries from hypothetical population means.

```r
stopifnot(nrow(d) == 98,
          identical(levels(d$type), c("bc", "prof", "wc")))
ch8a_groups <- data.frame(
  type = factor(c("bc", "prof", "wc"), levels = levels(d$type))
)
model.matrix(~type, data = ch8a_groups)
```

```
  (Intercept) typeprof typewc
1           1        0      0
2           1        1      0
3           1        0      1
attr(,"assign")
[1] 0 1 1
attr(,"contrasts")
attr(,"contrasts")$type
[1] "contr.treatment"
```

```r
ch8a_group_fit <- lm(income_k ~ type, data = d)
cbind(ch8a_groups,
      observed = as.numeric(tapply(d$income_k, d$type, mean)),
      fitted = predict(ch8a_group_fit, ch8a_groups))
```

```
  type observed fitted
1   bc    5.374  5.374
2 prof   10.559 10.559
3   wc    5.052  5.052
```

This group-only description is not the adjusted type comparison in *m2*. In *m2*, indicators compare fitted values at the same education, prestige, and percentage women. The factor is one term represented by two columns; it is neither an ordered numeric score nor one selected reference-category coefficient.

Releveling changes the intercept and indicator coefficients, not the fitted values, residuals, or a fixed substantive contrast:

```r
ch8a_relevelled <- d
ch8a_relevelled$type <- relevel(d$type, ref = "wc")
ch8a_ref_fit <- lm(
  income_k ~ education + prestige + women + type,
  data = ch8a_relevelled
)
max(abs(fitted(m2) - fitted(ch8a_ref_fit)))
```

```
[1] 1.066e-14
```

The result is floating-point noise. No new question has appeared merely because the coordinates changed. Centering continuous predictors also changes the intercept's reference profile without changing fitted values. It cannot create cross-type support.

A contrast needs coefficient covariance

For a declared linear contrast $\theta = c'\beta$, its estimate and estimated variance are

$$\hat{\theta} = c'\hat{\beta}, \qquad \widehat{\mathrm{Var}}(\hat{\theta}) = c'\widehat{V}c.$$

The SE is the square root of that variance. Consider professional minus white-collar in *m2*, at the same three continuous predictors. Relative to the coefficient order intercept, education, prestige, women, professional, white-collar, use $c = (0, 0, 0, 0, 1, -1)'$.

```r
ch8a_c <- c(0, 0, 0, 0, 1, -1)
ch8a_V <- vcov(m2)
ch8a_est <- sum(ch8a_c * coef(m2))
ch8a_var <- drop(ch8a_c %*% ch8a_V %*% ch8a_c)
ch8a_se <- sqrt(ch8a_var)
c(estimate = ch8a_est, variance = ch8a_var, SE = ch8a_se)
```

```
estimate variance       SE
 0.08886  1.16639  1.07999
```

Numerically, the estimate is $0.324193 - 0.235331 = 0.088863$ thousand dollars. Its variance includes the covariance:

$$2.140641 + 0.969780 - 2(0.972018) = 1.166385.$$

Thus $SE = \sqrt{1.166385} = 1.079993$, and the classical t_{92} interval is approximately $[-2.056, 2.234]$. Adding the two SEs, or adding variances while discarding covariance, gives a different and incorrect calculation. This interval is a working-model benchmark, not national sampling uncertainty. Nor does it overcome the limited observed cross-type support for the hypothetical same-covariate comparison.

Classical ANOVA and explicit block tests

For the group-only model, the null is equality of all three means, equivalently both type coefficients zero. The reduced model contains only an intercept. The full model has three coefficient columns. The classical statistic is

$$F = \frac{(RSS_{\text{reduced}} - RSS_{\text{full}})/2}{RSS_{\text{full}}/(98-3)}. \tag{3}$$

This is exactly the classical one-way ANOVA statistic: the same fitted group means, within-group RSS, two restrictions, denominator df 95, and null reference under independent normal common-variance errors. The equality is algebraic; it does not depend on randomized group membership.

```
ch8a_null <- lm(income_k ~ 1, data = d)
ch8a_nested <- anova(ch8a_null, ch8a_group_fit)
ch8a_aov <- aov(income_k ~ type, data = d)
c(max_fitted_difference = max(abs(
    fitted(ch8a_group_fit) - fitted(ch8a_aov))),
  regression_F = ch8a_nested$F[2],
  anova_F = summary(ch8a_aov)[[1]][["F value"]][1])
```

```
max_fitted_difference          regression_F               anova_F
                 0.00                 24.87                 24.87
```

An adjusted type test has different full-model dimensions. It compares *m2* with a model containing education, prestige, and women, giving two restrictions but denominator df $98-6 = 92$. Adding women and type jointly to *m1* instead tests three restrictions.

```
ch8a_no_type <- lm(
  income_k ~ education + prestige + women, data = d
)
ch8a_women_type <- anova(m1, m2)
ch8a_type <- anova(ch8a_no_type, m2)
```

Table 14

Two declared adjusted block comparisons.

Added	df	F	p
Women and type	3	11.509	<0.001
Type after women	2	0.032	0.969

The first comparison gives $F(3, 92) = 11.509$; the second gives $F(2, 92) = 0.032$, $p = .969$. Joint evidence for the first block does not establish that every added variable contributes. The second result is not proof that occupation types are meaningless.

Identical rows and the same outcome are necessary for this RSS comparison, but not sufficient: the models must be nested, their dimensions correct, and the reference-distribution conditions stated. Calling *anova(m2)* alone instead gives sequential sums of squares in formula order. Those are not order-free tests of every term. Welch and robust covariance tests are different procedures, not interchangeable versions of the classical formula.

Matrix geometry and general restrictions

Let X have n rows and p coefficient columns, including the intercept. Full column rank makes $X'X$ invertible. Differentiating $(y - X\beta)'(y - X\beta)$ gives the normal equations $X'(y - X\hat{\beta}) = 0$, hence

$$\hat{\beta} = (X'X)^{-1}X'y, \qquad \hat{y} = Hy, \qquad H = X(X'X)^{-1}X'.$$

The fitted vector is a projection onto the space spanned by the columns of X. Residuals are orthogonal to those columns. Changing a full-rank coding basis preserves that space and its fitted vector; adding a genuinely new column can enlarge it. A numerical solution is not itself a stochastic model or a substantive explanation.

```r
ch8a_X <- model.matrix(m2)
ch8a_beta <- solve(crossprod(ch8a_X),
                   crossprod(ch8a_X, d$income_k))
c(columns = ncol(ch8a_X), rank = qr(ch8a_X)$rank,
  max_coefficient_difference = max(abs(ch8a_beta - coef(m2))),
  max_normal_equation_error = max(abs(
    crossprod(ch8a_X, residuals(m2)))))
```

```
                   columns                       rank 
                 6.000e+00                  6.000e+00 
max_coefficient_difference  max_normal_equation_error 
                 2.057e-13                  5.969e-13 
```

R's *lm()* uses a numerically preferable QR calculation rather than requiring users to form an inverse. The displayed matrix equation explains the estimator, not a recommendation to replace its numerical implementation (Fox, 2016; Fox & Weisberg, 2019).

Under the classical variance model, $\widehat{V} = s^2(X'X)^{-1}$ with $s^2 = RSS/(n-p)$. For q independent linear restrictions $R\beta = r$, the classical statistic is

$$F = \frac{(R\hat{\beta} - r)'[R(X'X)^{-1}R']^{-1}(R\hat{\beta} - r)}{qs^2}.$$

It has the $F_{q,n-p}$ reference under the stated normal linear model. Choosing rows of R that select the two type coefficients recovers the adjusted type-block test. A robust replacement for the covariance requires its own approximate reference, not a claim of the same exact finite-sample result.

Another view of a partial slope removes the fitted linear relations with the other predictors from both income and education. If the remaining vectors are r_Y and r_E, then $b_E = r'_E r_Y/(r'_E r_E)$.

```r
ch8a_re <- residuals(lm(
  education ~ prestige + women + type, data = d
))
ch8a_ry <- residuals(lm(
  income_k ~ prestige + women + type, data = d
))
c(partial_slope = drop(
    crossprod(ch8a_re, ch8a_ry) / crossprod(ch8a_re)),
  partial_correlation = cor(ch8a_re, ch8a_ry))
```

```
      partial_slope partial_correlation
            0.11643             0.04406
```

The slope reproduces 0.116434; the partial correlation is 0.044063. This is linear residualization, not actual matching or randomization. A component-plus-residual plot in Section 4 is a different display: it adds a fitted component back to the outcome residual.

Extended practice: coding and known processes (Integrative)

1. Recover fitted *m2* values for the three types at education 11, prestige 50, and women 30. Treat this as algebra, then evaluate whether all three combinations have observed support. Relevel type to white-collar and verify a fixed fitted contrast is unchanged.
2. Reproduce the repeated-sample process at $n = 150$. Change exactly one of the correlation's sign, its value to zero, or the omitted variable's coefficient to zero. Derive both targets, run at least 1,000 repetitions with a recorded seed, and compare bias and SD relative to each target. An optional harder version uses separated predictor clusters and explicitly recalculates their covariance.

Appendix B: Technical Reproduction and Sensitivity

The numbered route now carries the ordinary support, diagnostic, standardization, and targeted-sensitivity lessons. This appendix retains full computational reproduction and mathematical detail. It uses *d*, *m0*, *m1*, and *m2* without changing their source rows. Classical intervals are conditional-model benchmarks; technical refinements do not create missing design, measurement, independence, or transport warrants.

Reproduce validation and inspect resplit sensitivity

The following complete function implements the main comparison. Both rules use the same training rows and the same held-out records. It checks factor support and rank before prediction; it does not use held-out outcomes to select variables or transform predictors.

```r
cv_compare <- function(data, fold) {
  stopifnot(length(fold) == nrow(data), !anyNA(fold),
            identical(levels(data$type), c("bc", "prof", "wc")))
  vars <- c("income_k", "education", "prestige", "women", "type")
  stopifnot(all(complete.cases(data[, vars])))
  pred <- matrix(NA_real_, nrow(data), 2,
                 dimnames = list(NULL, c("Education-only", "Full")))
  for (k in sort(unique(fold))) {
    train <- data[fold != k, ]
    test <- data[fold == k, ]
    stopifnot(all(table(train$type) > 0))
    simple <- lm(income_k ~ education, data = train)
    full <- lm(income_k ~ education + prestige + women + type,
               data = train)
    stopifnot(simple$rank == 2, full$rank == 6,
              nobs(simple) == nrow(train),
              nobs(full) == nrow(train))
    pred[fold == k, 1] <- predict(simple, newdata = test)
    pred[fold == k, 2] <- predict(full, newdata = test)
  }
  stopifnot(all(is.finite(pred)))
  errors <- abs(pred - data$income_k)
  mae <- colMeans(errors)
  list(predictions = pred, absolute_errors = errors, MAE = mae,
       difference = unname(mae[2] - mae[1]))
}
```

With the function defined, recreate the stated folds and collect the errors.

```r
RNGkind("Mersenne-Twister", "Inversion", "Rejection")
set.seed(80806)
ch8b_fold <- sample(rep(seq_len(10), length.out = nrow(d)))
ch8b_primary <- cv_compare(d, ch8b_fold)
ch8b_primary$MAE
```

```
Education-only           Full
         2.403          1.541
```

```r
ch8b_primary$difference
```

```
[1] -0.8619
```

The primary MAEs reproduce 2.403130 and 1.541238, with full minus education-only difference −0.861892 thousand dollars. The function averages over cases. An unweighted average of ten fold means would give the two nine-case folds too much weight. The case-weighted target mixes training sizes 88 and 89 with weights 80/98 and 18/98.

The fixed unit conversion *income/1000* is not a learned preprocessing step. Any future learned transformation, imputation, or variable selection would have to occur inside each training fold. The complete record restriction instead defines this exercise's eligibility; it does not make missingness harmless for a wider population.

The predeclared sensitivity uses 200 new random fold allocations, without modifying either rule after looking at their errors:

```r
set.seed(80807)
ch8b_resplits <- t(replicate(200, {
  f <- sample(rep(seq_len(10), length.out = nrow(d)))
  ans <- cv_compare(d, f)
  c(Education = ans$MAE[1], Full = ans$MAE[2],
    Difference = ans$difference)
}))
```

Table 15
Fold-allocation sensitivity, not confidence limits.

Quantity	Mean	Minimum	Maximum
Education-only MAE	2.420	2.381	2.490
Full MAE	1.589	1.527	1.671
Full minus simple	-0.831	-0.914	-0.741

The mean difference is -0.831149, with range -0.913974 to -0.740997; all 200 allocations favor the full rule. This describes resplit sensitivity on the same data. The overlapping training sets, reused outcomes, and repeated predictions are not independent new samples. Neither that range nor a standard error computed by treating fold means as independent is a sampling interval for future risk.

The internal result concerns learning rules trained at the stated sizes, under the same-source assumptions. It is not an independent test of the final all-98 fitted model. It cannot validate modern occupations, another country, individual workers, or unseen types. Nor does it establish which added predictor is responsible: prestige, women, and type were added together. This remains retrospective teaching reanalysis, not a historical preregistration claim.

Cook's distance and full deletion accounting

The main route interprets residual, leverage, and influence views and shows the education-slope sensitivity. This section supplies the corresponding Cook's distance formula and complete coefficient accounting.

With p coefficient columns including the intercept, leverage is $h_{ii} = x_i'(X'X)^{-1}x_i$. Cook's distance can be calculated as

$$D_i = \frac{e_i^2}{ps^2}\frac{h_{ii}}{(1-h_{ii})^2} = \frac{(\hat{\beta}-\hat{\beta}_{(-i)})'X'X(\hat{\beta}-\hat{\beta}_{(-i)})}{ps^2}.$$

The second expression measures a coefficient change using the design matrix's metric, not an unweighted sum of squared coefficients with incompatible units.

Table 16
The five largest Cook's distances in the full fit.

occupation	residual	leverage	cooks
general.managers	14.206	0.077	0.447
physicians	11.009	0.079	0.272
ministers	-7.762	0.040	0.063
lawyers	5.385	0.063	0.050
surveyors	-4.901	0.063	0.042

The record for general managers has the largest Cook's distance, approximately 0.447. The main route explains why this observation is inspected and retained.

Table 17
Coefficient sensitivity to the most influential record.

Term	Full	Omit_one	Difference
(Intercept)	0.421	-1.526	-1.947
education	0.116	0.437	0.321
prestige	0.141	0.118	-0.023
women	-0.054	-0.050	0.004
typeprof	0.324	-1.299	-1.623
typewc	0.235	-0.597	-0.832

The type coordinates also move. The table records a diagnostic perturbation of the fit, not six new research questions or evidence that the reduced sample is preferable.

HC3 changes covariance, not the records or target

For independent observations, the HC3 covariance estimator is

$$\widehat{V}_{\text{HC3}} = (X'X)^{-1}\left\{\sum_i x_i x_i' \frac{e_i^2}{(1-h_{ii})^2}\right\}(X'X)^{-1}.$$

The outer factors are the familiar design component. The middle component allows different residual scales and adjusts for leverage (Long & Ervin, 2000; Zeileis, 2004). It leaves the OLS coefficients unchanged.

```r
ch8b_Vhc3 <- sandwich::vcovHC(m2, type = "HC3")
ch8b_X <- model.matrix(m2)
ch8b_inverse <- solve(crossprod(ch8b_X))
ch8b_adjusted_e <- residuals(m2) / (1 - hatvalues(m2))
ch8b_manual <- ch8b_inverse %*%
  crossprod(ch8b_X * ch8b_adjusted_e) %*% ch8b_inverse
max(abs(ch8b_manual - ch8b_Vhc3))
```

```
[1] 1.406e-12
```

Table 18

Coefficient estimates and two covariance summaries.

Term	Estimate	Classical_SE	HC3_SE
(Intercept)	0.421	1.957	2.774
education	0.116	0.275	0.382
prestige	0.141	0.035	0.039
women	-0.054	0.010	0.008
typeprof	0.324	1.463	2.093
typewc	0.235	0.985	1.055

Education's HC3 SE is 0.382333 rather than 0.275226. Using t_{92} as an approximate reference gives $[-0.642912, 0.875779]$. This is not an exact finite-sample HC3 coverage result. Women's HC3 SE is smaller than its classical counterpart: robust covariance need not widen every interval.

For a maintained linear conditional mean, HC3 addresses unequal variance under independent observations. For a random-design best-linear-projection target, a suitable sandwich covariance can support large-sample projection inference without making the mean exactly linear. Neither interpretation repairs measurement error, dependence, selection, causal identification, or absent joint support.

VIF describes coefficient precision

For a single predictor column, classical conditional variance can be written

$$\mathrm{Var}(\hat{\beta}_j \mid X) = \frac{\sigma^2}{\sum_i (X_{ij} - \bar{X}_j)^2} \frac{1}{1 - R_j^2}.$$

Here R_j^2 comes from regressing that predictor on the remaining columns, including the intercept. The variance inflation factor is $VIF_j = 1/(1 - R_j^2)$; its square root is an SE multiplier, holding the residual variance and focal predictor variation fixed (Kutner et al., 2005).

```
ch8b_aux <- lm(education ~ prestige + women + type, data = d)
ch8b_r2 <- summary(ch8b_aux)$r.squared
c(auxiliary_R2 = ch8b_r2, VIF = 1 / (1 - ch8b_r2),
  SE_multiplier = sqrt(1 / (1 - ch8b_r2)))
```

```
 auxiliary_R2           VIF SE_multiplier
       0.8765        8.0968        2.8455
```

```
car::vif(m2)
```

```
           GVIF Df GVIF^(1/(2*Df))
education 8.097  1           2.845
prestige  5.170  1           2.274
women     1.301  1           1.140
type      7.700  2           1.666
```

Education gives $R_j^2 = 0.876495$, VIF 8.096833, and SE multiplier 2.845494. The predictor correlation between education and prestige is 0.866480, but VIF uses all the other columns, not just that pair.

The two-column type term has a generalized VIF. Its raw GVIF is 7.700131; the dimension-adjusted value $GVIF^{1/(2\,df)} = 1.665806$ is on a comparable scale to the square root of a one-column VIF (Fox & Weisberg, 2019). Do not compare a raw two-column GVIF with an ordinary one-column VIF as if their dimensions matched.

Exact collinearity creates rank deficiency. Near collinearity leaves coefficients estimable but potentially imprecise. Under the maintained zero-conditional-mean model it does not itself bias OLS. A cutoff of 5 or 10 neither establishes all assumptions nor justifies removing a variable required by the question. Measurement problems also need their own evidence: the familiar simple attenuation story requires classical error assumptions and is not a universal rule for every coefficient in a multiple-predictor model.

Changed outcome scales and in-sample comparison criteria

The core already examines a prestige-quadratic sensitivity while preserving the income outcome. A log-income model changes the outcome scale and therefore requires an explicitly changed question.

```r
ch8b_log <- lm(
  log(income) ~ education + prestige + women + type, data = d
)
100 * (exp(coef(ch8b_log)[c("education", "prestige", "women")]) - 1)
```

```
education  prestige     women 
   0.7661    2.0965   -0.8430 
```

The log-income model asks about additive differences in expected log income, or multiplicative differences in geometric means under that model. A one-unit contrast gives $100(\exp(\hat{\beta}) - 1)$ percent, approximately $100\hat{\beta}$ for small coefficients. It is not automatically an arithmetic-mean-income percentage effect. A changed outcome requires a named changed question, not a claim that the data have merely been cleaned.

Table 19

Log-model contrasts on a geometric-mean percent scale.

Term	Log_coefficient	Percent	Lower	Upper
education	0.008	0.766	-4.204	5.994
prestige	0.021	2.096	1.435	2.762
women	-0.008	-0.843	-1.019	-0.667

These intervals transform classical log-scale coefficient intervals. Their interpretation remains conditional on the log-scale working model; transformation does not supply a causal or population warrant.

For comparable Gaussian models on the same observations and outcome,

$$AIC = -2 \log \hat{L} + 2k, \qquad BIC = -2 \log \hat{L} + k \log n.$$

Here k includes regression coefficients **and** estimated residual variance. Smaller values are preferred under each criterion, but neither is the observed held-out MAE (Akaike, 1974; Schwarz, 1978).

Table 20
In-sample summaries for comparable income models.

	n	k	R2	Adjusted_R2	AIC	BIC
m0	98	3	0.330	0.323	526.5	534.3
m1	98	4	0.500	0.489	499.8	510.2
m2	98	7	0.636	0.616	474.6	492.7

The raw AIC/BIC values for income and log income must not be compared as if they used the same response density. A scale change changes the likelihood's reference measure; a valid common-scale comparison would need the corresponding transformation adjustment. The simpler lesson here is to preserve the outcome and question in each comparison.

No residual graph establishes independence. The occupations share one historical labour market, and metadata do not establish a probability sample of a wider population. For students within schools or repeated observations, the design may require dependence-aware uncertainty. HC3 alone is not such a repair.

The practical response is consequential rather than pass/fail: retain the named comparison, report its form and influence sensitivities, distinguish classical from robust uncertainty, and bound interpretation by measurement, support, and source. Causal adjustment-set rules about confounders, mediators, and colliders apply to a declared causal target; they are not universal requirements for association or prediction.

Extended practice: reproduction and critique (Integrative)

1. Compute the white-collar conditional profile (11, 45, 50) and the empirical-profile standardized value, with appropriately labelled intervals. Name each target and assess support. Propose different weights and explain which part of the target changes.
2. Rebuild the 102-to-98 audit and three-model sequence from a clean session. Preserve names, units, type levels, and source row order. Add one supported display and a bounded occupation-level report.
3. A school-level analysis reports a slope 0.40, classical SE 0.18, and HC3 SE 0.31. Its residual smooth curves, spread increases, deleting one influential school changes the slope to -0.05, and sectors have little prior-score overlap. Assess the claim: "The causal effect is robust because VIFs are below 10; deleting the unusual school will make the model reliable." Match each finding to its consequence and propose a transparent sensitivity.
4. Choose another documented public dataset with a continuous outcome, two continuous predictors, and a factor with at least three levels. Declare an additive question and analytic sample before fitting. Document source, licence, units, missingness, coding, support, conditional interpretation, a block comparison, checks, and one sensitivity. Read the reported reasons and answer, then separately evaluate them. Do not require every

changed estimator to imply a changed scientific estimand. Record versions and stochastic seeds.

References

Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, *19*(6), 716–723. https://doi.org/10.1109/TAC.1974.1100705

Chetty, R., Friedman, J. N., Hilger, N., Saez, E., Schanzenbach, D. W., & Yagan, D. (2011). How does your kindergarten classroom affect your earnings? Evidence from Project STAR. *The Quarterly Journal of Economics*, *126*(4), 1593–1660.

Cook, R. D. (1977). Detection of influential observation in linear regression. *Technometrics*, *19*(1), 15–18. https://doi.org/10.2307/1268249

Finn, J. D., & Achilles, C. M. (1990). Answers and questions about class size: A statewide experiment. *American Educational Research Journal*, *27*(3), 557–577.

Fox, J. (2016). *Applied regression analysis and generalized linear models* (3rd ed.). SAGE Publications.

Fox, J., & Weisberg, S. (2019). *An R companion to applied regression* (3rd ed.). SAGE Publications. https://www.john-fox.ca/Companion/index.html

Hanushek, E. A. (1999). Some findings from an independent investigation of the Tennessee STAR experiment and from other investigations of class size effects. *Educational Evaluation and Policy Analysis*, *21*(2), 143–163.

Kleiber, C., & Zeileis, A. (2008). *Applied econometrics with R*. Springer. https://doi.org/10.1007/978-0-387-77318-6

Krueger, A. B. (1999). Experimental estimates of education production functions. *The Quarterly Journal of Economics*, *114*(2), 497–532.

Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2005). *Applied linear statistical models* (5th ed.). McGraw-Hill/Irwin.

Long, J. S., & Ervin, L. H. (2000). Using heteroscedasticity consistent standard errors in the linear regression model. *The American Statistician*, *54*(3), 217–224. https://doi.org/10.1080/00031305.2000.10474549

Mosteller, F. (1995). The Tennessee study of class size in the early school grades. *The Future of Children*, *5*(2), 113–127.

Nye, B., Hedges, L. V., & Konstantopoulos, S. (2000). The effects of small classes on academic achievement: The results of the Tennessee class size experiment. *American Educational Research Journal*, *37*(1), 123–151.

Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics*, *6*(2), 461–464. https://doi.org/10.1214/aos/1176344136

Stock, J. H., & Watson, M. W. (2007). *Introduction to econometrics* (2nd ed.). Addison Wesley.

Zeileis, A. (2004). Econometric computing with HC and HAC covariance matrix estimators. *Journal of Statistical Software*, *11*(10), 1–17. https://doi.org/10.18637/jss.v011.i10